HITACHI
Hitachi Rail SpA

OBS	FUNCTIONAL REQUIREMENT SPECIFICATION
DEPARTMENT	DOCUMENT TYPE

PROJECT NAME

ETR1000 VZI R1

DOCUMENT TITLE

SFRS - FIRE PROTECTION SYSTEM

NONDISCLOSURE CLAUSE

Copyright reserved.

È vietato consegnare a terzi o riprodurre questo documento, utilizzarne il contenuto o renderlo comunque noto a terzi senza esplicita autorizzazione. Ogni infrazione comporta il risarcimento dei danni subiti. Sono riservati tutti i diritti derivanti dalla concessione di brevetti per invenzioni industriali di utilità o di brevetti per modelli ornamentali.

The reproduction, distribution and utilization of this document as well as the communication of its contents to other without express authorization is prohibited. Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or design.

HITACHI Hitachi Rail SpA		DATE	NAME	DOCUMENT NUMBER	FA020025419
	CREATED	15-04-2024	G. Guidi	PDM REVISION	02
	CHECKED	15-04-2024	S. Tesi	DOORS MODULE NAME	FRS
	APPROVED	15-04-2024	S. Di Lella	DOORS BASELINE	0.1

Revision History

Rev	Date	Author	Revision Description
00	28-11-2023	G. Guidi	First Issue – Based on document EC01P008884B (Baseline 1.3)
01	29-03-2024	G. Guidi	PARAGRAPHS CHANGES: - New Paragraphs: 2.2.2, 3.4.1.1.4, 3.4.1.1.5. - Modified Paragraphs: 2.2, 3.1.1.2.2, 3.1.1.2.3, 3.2.1.1, 3.2.1.3, 3.2.1.4, 3.2.1.5, 3.3.1, 3.3.1.1, 3.6.1, 3.3.1.1.3. - Deleted Paragraphs: - REQUIREMENTS CHANGES: - New Requirements: 2F_0y.0z.Zefiro-Europe.CONCEPT.HR41, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR44, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR45, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR42, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR23, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR46, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR47, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR48, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR49, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR50, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR51, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR57, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR58, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR59, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR54, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR60, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR61, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR79, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR62, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR76, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR77, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR56, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR80, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR63, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR64, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR65, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR68, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR69, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR70, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR71. - Modified Requirements: 2F_0y.0z.Zefiro-Europe.CONCEPT.HR01, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR09, 2F_0y.0z.Zefiro-Europe.CONCEPT.160, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR04, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR05, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR10, 2F_0y.0z.Zefiro-Europe.CONCEPT.236, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR13, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR14, 2F_0y.0z.Zefiro-Europe.CONCEPT.86, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR43, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR22, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR25, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR26, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR28, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR36, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR38, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR39, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR40, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR52, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR53, 2F_0y.0z.Zefiro-Europe.CONCEPT.77, 2F_0y.0z.Zefiro-Europe.CONCEPT.79, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR87, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR24, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR35, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR72, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR73, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR74, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR75, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR76, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR23, 2F_0y.0z.Zefiro-Europe.CONCEPT.116. - Deleted Requirements: 2F_0y.0z.Zefiro-Europe.CONCEPT.HR23, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR88, 2F_0y.0z.Zefiro-Europe.CONCEPT.238, 2F_0y.0z.Zefiro-Europe.CONCEPT.239, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR240, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR241, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR242, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR26, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR27, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR28, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR30, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR31, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR49, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR50, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR51.
02	15-04-2024	G. Guidi	PARAGRAPHS CHANGES: - New Paragraphs: - - Modified Paragraphs: 3.3.1 - Deleted Paragraphs: - REQUIREMENTS CHANGES: - New Requirements: 2F_0y.0z.Zefiro-Europe.CONCEPT.HR78, 2F_0y.0z.Zefiro-Europe.CONCEPT.HR79 - Modified Requirements: - - Deleted Requirements: -

Table of contents

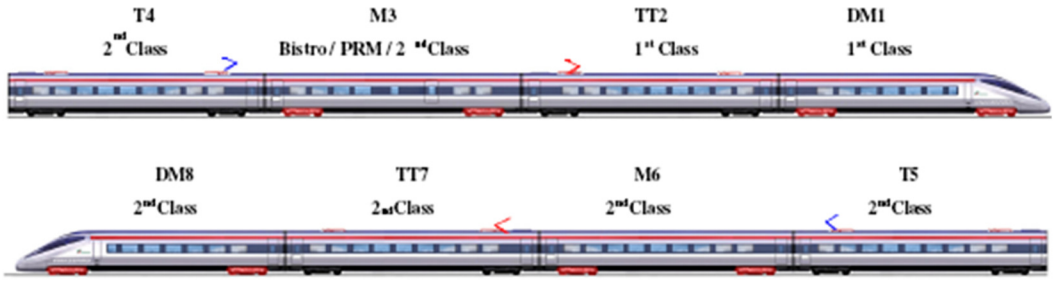
1. Introduction 4
1.1 Purpose 4
1.2 Train Configuration 4
1.3 Definitions and Acronyms 4
2. General 5
2.1 System Configuration 5
2.2 Man Machine Interface 9
2.2.1 Communication with TCMS of the FPS diagnostic status 9
2.2.2 Fault table 10
3. Functional Requirements 14
3.1 System Configuration 14
3.1.1 Functional Design 14
3.1.2 Requirements 24
3.2 Supervision of the fire detection system 25
3.2.1 Function Design 25
3.2.2 Requirements 27
3.3 Driver interface 30
3.3.1 Smoke/heat alarm 30
3.3.2 Mute Function 38
3.3.3 Reset Function 39
3.3.4 Display detectors status on IDU 41
3.4 Fire Extinguisher 43
3.4.1 Function Design 43
3.4.2 Requirements 55
3.4.2 Interrupt fire extinguisher 56
3.5 Maintenance/Test Mode of fire detection system 57
3.5.1 Function design 57
3.5.2 Requirements 57
3.6 System Fault Managment 58
3.6.1 Function design 58
3.7 Events 60
3.7.1 Function design 60
3.7.2 Requirements 60
4. Signal Interface 61
5. References 62

1. Introduction

1.1 Purpose

This document is a function specification for the control of the Fire Detection and Extinguishing system for ETR1000 VZI R1 for customer Trenitalia.

1.2 Train Configuration



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.3 Definitions and Acronyms

Acronym	Meaning
CD	Circuit Diagram
DV1	WSU activation valve
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
FCL	Fire Control Line
FDU	Fire Detection Unit
FPS	Fire Protection System
FSK	Frequency Shift Keying
HV	High Voltage
HVAC	Heat Ventilation and Air Conditioning
HWCI	Hardware Configuration Item
HW HMI	Hardware Human Machine Interface, e.g. buttons, swiches and indication lamps.
IDU	Interface Display Unit
LHD	Linear Heat Detector
MCL	Main Control Line
NC	Normally Close
NO	Normally Open
PB	Push Button
PSU	Power Supply Unit
PTE	Portable Test Equipment
RCL	Rack I/O Communication Line
RIO	Rack Input/Output
STD	Smoke Temperature Device
TCMS	Train Control and Monitoring System
TDV	Test drain Valve
TOD	Train Operator Display
TRDP	Train Real-time Data Protocol
TU	Traction Unit
VCB	Vacuum Circuit Breaker
WSU	Water Supply Unit

2. General

2.1 System Configuration

The Fire Protection System (FPS) is the result of integrating 3 different parts:

- Fire detection;
- Water mist fire suppression in passenger/staff areas;
- Aerosol extinguishing in technical areas.

Fire detection is performed everywhere in passenger areas (passengers saloons, vestibules, toilets and bistrò) and staff areas (staff toilet, driver cabs and electronic rooms). In these zones the fire is detected by using point detectors capable of detecting the presence of smoke and the temperature increase.

Fire suppression in passenger/staff areas is guaranteed (with the exception of the driver cabs) through the installation of a WSU, while in technical areas it is guarendeed through the installation of aerosol cartriges.

The main functions of the Fire Protection System are summarized in the following points:

- Fire detection in passenger areas and staff areas;
- Fire detection in technical areas;
- Control and monitoring of the WSU extinguishing system for passenger areas;
- Activation of the extinguishing systems in the technical compartments;
- Indication of the fire ALARM status and FAULT status;
- Management of Alarm and Fault signals from/to the coupled train;
- Communication with TCMS of the FPS diagnostic status.

Fire Protection System is composed by:

- Two (2) Fire Detection Unit (FDU): the main control units of the system. One FDU is installed in DM1 and the other is installed in DM8. FDUs perform the following main actions:
    - Interface with STD smoke sensors installed in the passenger areas and in the staff areas;
    - Alarm condition detection in the passenger area and in the staff areas;
    - Fire extinguisher activation command inside the passenger areas and staff areas in case of ALARM detection (with the exception of crew cabins and electronic cabinets);
    - Control and monitoring of the “Water Supply Unit” (WSU) installed in DM1;
    - ALARM signals and FAULT signals management;
    - ALARM signals and FAULT signals management from/to the coupled train;
    - Interface with the control units RIO installed under-frame to supervise and protect the technical compartments;
    - Diagnostic interface with TCMS via TRDP protocol.
- Eight (8) Rack Input/Output (RIO): local control devices, installed one for each car, which performs the following actions:
    - Acquire the status of the LHD cables mounted in the relative technical compartments;
    - Acquire the status of the aerosols present in the respective technical compartments;

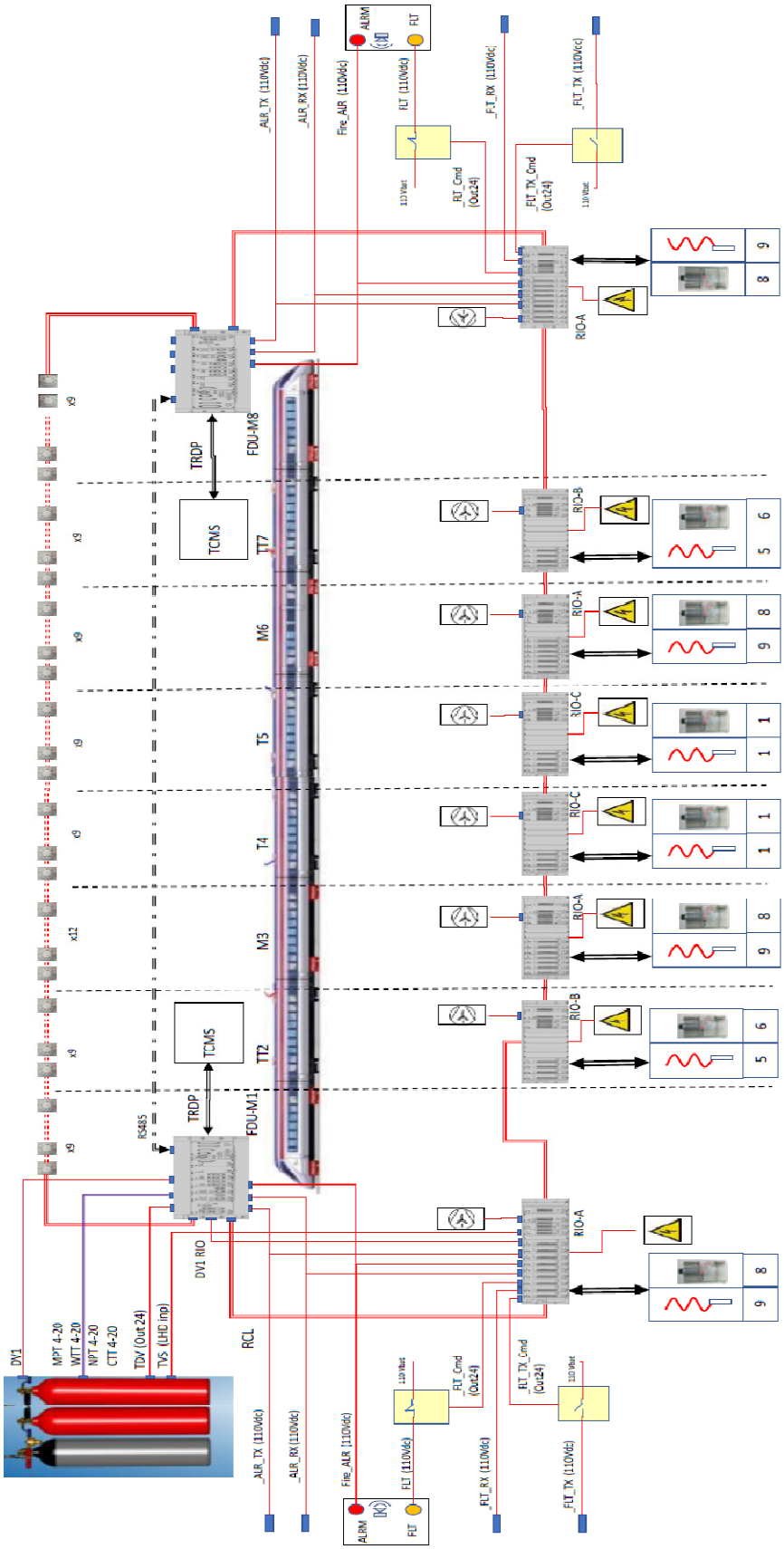
- Report diagnostics of the LHD cables and aerosols to the FDU;
- Detect the presence of an alarm condition in a technical compartment through the short-circuit detection of the LHD cable;
- Activate the aerosol(s) to protect the technical compartment under alarm based on command from the FDU;
- Activate the security hardwired-lines dedicated to:
  - Deactivation of the HVACs of the coach in alarm, on command from the FDU;
  - Deactivation (Shutdown) of the Power Supply in the monitored coach, on command from the FDU, in case of fire ALARM detected in a technical area;
  - Alarm's line power supply in order to pilot the lamp on driver desk on command of the FDU in case of fire alarm only DM1;
  - Transmission of the fire ALARM status to the Traction Unit (TU) based on command from the FDU in the event of a fire alarm;
  - Transmission of the FAULT status to the Traction Unit (TU) based on command from the FDU in the event of a failure.
- Acquisition of the status of the TVS valve sensors and communicate it on request from the FDU control unit.

- Seventy-two (72) Smoke and Temperature Detectors (STD): installed along the train compartments. They control the presence of fire aboard through the following parameters: Smoke, Temperature gradient and Temperature.
- Six-teen (48) Linear Heat Detectors (LHD): installed in each Technical Compartment monitored by Fire Protection System (Traction Converters, Transformer Control Boxes, AC High Voltage Boxes, DC Main Circuit breakers, Transformer Cooler, Charging Boxes, Distribution Boxes). These cables are able to detect the persistence of a temperature higher than a predetermined threshold characteristic of the type of detector.
- One (1) Water Supply Units (WSU): The WSU is installed in DM1, which contains the water mist needed for fire suppression in passenger areas (including vestibules, toilets and bistrò) and staff areas (except for the driver cabs where no sprinklers are installed).

  The suppression system is designed at train level: the water/gas cylinders will be installed only in coach DM1 and there will be a pipe network with a main branch, going along the whole train and connecting the suitable derivations to the sprinklers, in order to deliver the water mist where the fire is detected.
- Ninety-six (96) Sprinklers: Each Sprinkler will be connected to the related adapter and will be protected with a cap. The nozzle head contains a heat sensitive bulb that breaks in case of high temperature and discharge the water mist only locally. The suppression system is commanded by the detection system that activates the WSU unit in case of fire.
- Fourty-six (46) FireKloud Aerosol (FKA): The extinguishing system in technical areas is performed by using aerosol cartridges installed in each enclosure. The extinguishing system is commanded by the detection system that activates the aerosol(s) of the enclosure where fire has been detected (after a pre definite delay time in order to let possible the fan ventilation stop).

The interface between FDUs and smoke sensors, as well as their power supply is made through a shielded two-wire cable using the digital protocol licensed from MicrodataDue (FCL).
The interface between FDUs and RIOs takes place via a two-wire communication bus shielded using the FSK serial protocol (RCL).
The two FDU control units, installed in boxes DM1 and DM8, communicate with each other by means of a serial RS485 (MCL).
The interface between FDUs and the on-board TCMS is made on an Ethernet line with TRDP protocol.
Moreover, the fire protection system is connected to the driver desks via HW signals in order to light a lamp in case of a fire alarm and with the HVAC systems via HW signals in order to shut-down ventilation and close external shutters of the HVAC units of the coach where the fire has been detected.

Following the functional diagram of the system:


ALRM
FLT
_ALR_TX (110Vdc)
_ALR_RX (110Vdc)
Fire_ALR (110Vdc)
FLT (110Vdc)
_FLT_RX (110Vdc)
_FLT_TX (110Vdc)
FLT_Cmd (Out24)
FLT_TX_Cmd (Out24)
110 Vdc
RIO-A
RIO-B
RIO-C
FDU-M8
FDU-M11
TRDP
TCMS
RS485
TT7
M6
T5
T4
M3
TT2
x9
x12
DV1
MPT 4-20
WTT 4-20
NPT 4-20
CTT 4-20
TDV (Out 24)
TVS (LHD inp)
DV1 RIO
RCL

2.2 Man Machine Interface

Driver Interface on cab desk:

- FIRE ALARM lamp (Red Light);
- FAULT lamp, suitable for system DEGRADED/OUT OF ORDER.

User Interface on FDU:

- FIRE ALARM RESET push-button (RESET)
- STD sensor disabling push-button (DISABLE)
- TEST/OPER selector (TST/OPER)
- POWER SUPPLY indicators (led 1, led 2)
- MASTER/SLAVE indicators (led 3, led 4)
- SAFE/FAULT indicators (led 5, led 6)
- PRE-ALARM/ALARM indicators (led 7, led 8)

Passenger Interface: N.A.

Interface for Diagnostic Purpose: TOD.

2.2.1 Communication with TCMS of the FPS diagnostic status

The FPS system shall communicate its status and its diagnostics with the on-board train system TCMS via an Ethernet port.
The communication protocol used is the TRDP standard.

2.2.2 Fault table

In the table below is summarized the faults foreseen:

Id Fault	FDS	FDU Master	FDU Slave	UNI 11565	Safety	TCMS	Lampada Fault	Comportamento FDU
1	Perdita comunicazione con il TCMS	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Cambio mastership 2. Blocco negoziazione
2	Perdita comunicazione con il TCMS	x	x	MINOR	NON CRITICO	DEGRADATO	OFF	1. Proseguimento operazioni
3	Guasto critico interno alla FDU (FDU in Fault)	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Cambio mastership 2. Blocco negoziazione
4	Guasto critico interno alla FDU (FDU in Fault)	x	x	CRITICAL	CRITICO	FUORI SERVIZIO	ON	1. Blocco operazioni
5	Perdita totale linea STD (FCL)	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Cambio mastership 2. Blocco negoziazione
6	Perdita totale linea STD (FCL)	x	x	MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
7	Perdita totale linea RIO (RCL)	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Cambio mastership 2. Blocco negoziazione
8	Perdita totale linea RIO (RCL)	x	x	MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
9	Perdita linea RS485 (MCL)	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Blocco negoziazione
10	Perdita entrambe le linee RCL, FCL	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Cambio mastership 2. Blocco negoziazione
11	Perdita entrambe le linee RCL, FCL	x	x	CRITICAL	CRITICO	FUORI SERVIZIO	ON	1. Blocco operazioni
12	Perdita linea RS485 (MCL) e una delle due FCL, RCL.	x		MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
13	Perdita di un sensore STD in una delle Aree passeggeri presidiata soltanto da uno o due sensori	x		MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni
14	Perdita di due sensori STD in una delle Aree passeggeri presidiata da più di due sensori	x		MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni
15	Perdita di un sensore STD in una delle Aree passeggeri presidiata da più di due sensori.	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
16	Rilevazione di una condizione di Allarme di un sensore STD in una delle Aree Passeggeri presidiate da soltanto uno o due sensori, immediatamente all'uscita dalla fase di bootstrap della FDU Master	x		MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni 2. Blocco negoziazione mastership
17	Rilevazione di una condizione di Allarme di due sensori STD appartenenti ad una qualsiasi delle Aree Passeggeri, immediatamente all'uscita dalla fase di bootstrap della FDU Master	x		MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni 2. Blocco negoziazione mastership
18	Rilevazione di una condizione di Allarme di un sensore STD in una delle Aree Passeggeri presidiate da più di due sensori, immediatamente all'uscita dalla fase di bootstrap della FDU Master	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
19	Perdita di rilevazione di un LHC a presidio di uno dei comparti tecnici causato da un guasto sul cavo (condizione open).	x		MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni

20	Perdita di rilevazione di un LHC a presidio di uno dei comparti tecnici causato da un guasto nella circuiteria di rilevazione.	x		MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni
21	Rilevazione di una condizione di Allarme di un LHC a presidio di uno dei comparti tecnici immediatamente all'uscita dalla fase di bootstrap della FDU Master (Il cavo viene considerato guasto).	x		MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni 2. Blocco negoziazione mastership
22	Anomalia sulla diagnostica di un Aerosol a presidio di uno dei comparti tecnici (anomalia sulla scheda O24)	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
23	Rilevazione circuito aperto sulla diagnostica di un Aerosol a presidio di uno dei comparti tecnici.	x		MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni
24	Rilevazione corto circuito sulla diagnostica di un Aerosol a presidio di uno dei comparti tecnici.	x		MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni
25	Interruzione linea STD (FCL)	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni con condivisione linea
26	Interruzione linea FSK (RCL)	x		MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni con condivisione linea
27	Interruzione simultanea linee STD (FCL) e FSK (RCL)	x		CRITICAL	CRITICO	FUORI SERVIZIO	ON	1. Blocco operazioni
28	Guasto sonda di temperatura all'interno del WSU (WTT)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
29	Guasto sonda di temperatura delle bombole di acqua (CTT)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
30	Guasto al trasduttore di pressione N2 (NPT)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
31	Guasto al trasduttore di pressione delle tubazioni (MPT)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
32	Valore di pressione nelle tubazioni superiore alla soglia massima stabilita per lo stato Stand-by in assenza di pre-attivazione			MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni
33	Valore di pressione nelle tubazioni superiore alla soglia massima stabilita per lo stato Stand-by dovuto alla condizione di pre-attivazione			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
34	Valore di pressione nelle tubazioni inferiore alla soglia minima stabilita durante la fase di scarica in stato di allarme			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
35	Valore di pressione nella bombola di azoto non coerente con la temperatura rilevata all'interno del WSU nello stato di Stand-by ed in assenza di pre-attivazione			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
36	Valore di pressione nella bombola di azoto inferiore alla soglia minima stabilita nello stato di Stand-by ed in assenza di pre-attivazione			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni

37	Valore di temperatura dell'acqua nei cilindri sotto il valore +1°C			MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni
38	Valore di temperatura ambiente dal TCMS non disponibile in assenza di pre-attivazione			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
39	Valore di temperatura ambiente dal TCMS non disponibile in condizioni di pre-attivazione (tubazioni piene di acqua)			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
40	Valore di temperatura ambiente indicata dal TCMS inferiore al valore +0°C +/- in condizione di pre-attivazione (tubazioni piene di acqua)			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
41	Valore di temperatura ambiente indicata dal TCMS inferiore al valore +0°C +/- senza pre-attivazione			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
42	Valvola TVS chiusa			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
43	Guasto del circuito diagnostico di rilevamento posizione valvola TVS			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
44	Perdita del circuito di attivazione dell'elettrovalvola DV1 nella FDU in DM1			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
45	Perdita del segnale di attivazione dell'elettrovalvola DV1 della RIO in DM1			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
46	Perdita di entrambi i canali di attivazione dell'elettrovalvola DV1 sulle centraline RIO ed FDU in DM1			MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni
47	Errore nel pilotaggio di una delle due linee di Allarme in DM1 (errore nel circuito di attivazione sulla RIO oppure errore nel circuito di attivazione sulla FDU)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
48	Errore nel pilotaggio di una delle due linee di Allarme in DM8 (errore nel circuito di attivazione sulla RIO oppure errore nel circuito di attivazione sulla FDU)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
49	Errore pilotaggio di entrambe le linee di Allarme in DM1 (errore nel circuito di attivazione sulla RIO e errore nel circuito di attivazione sulla FDU)			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
50	Errore pilotaggio di entrambe le linee di Allarme in DM8 (errore nel circuito di attivazione sulla RIO e errore nel circuito di attivazione sulla FDU)			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
51	Errore pilotaggio di una delle due linee Allarme verso il treno accoppiato in DM1 (errore nel circuito di attivazione sulla RIO oppure errore nel circuito di attivazione sulla FDU)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni

52	Errore pilotaggio di una delle due linee Allarme verso il treno accoppiato in DM8 (errore nel circuito di attivazione sulla RIO oppure errore nel circuito di attivazione sulla FDU)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
53	Errore pilotaggio di entrambe le linee di Allarme verso il treno accoppiato in DM1 (errore nel circuito di attivazione sulla RIO e errore nel circuito di attivazione sulla FDU)			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
54	Errore pilotaggio di entrambe le linee di Allarme verso il treno accoppiato in DM8 (errore nel circuito di attivazione sulla RIO e errore nel circuito di attivazione sulla FDU)			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
55	Errore pilotaggio di una delle due linee relè ridondate del Power Shutdown (guasto nel circuito di attivazione su una ORL)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
56	Errore pilotaggio entrambe le linee relè ridondate del Power Shutdown (guasto nel circuito di attivazione su entrambe le ORL) oppure mancanza di 110V			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
57	Errore pilotaggio di una delle due linee relè ridondate HVAC (guasto nel circuito di attivazione su una ORL)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
58	Errore pilotaggio entrambe le linee relè ridondate HVAC (guasto nel circuito di attivazione su entrambe le ORL) oppure mancanza di 110V			MAJOR	CRITICO	FUORI SERVIZIO	ON	1. Proseguimento operazioni
65	Errore nel pilotaggio della valvola TDV dalla FDU richiesto dal TCMS (rilettura dello stato del comando non coerente con il comando impartito)			MINOR	NON CRITICO	DIAGNOSTICA	OFF	1. Proseguimento operazioni
66	Guasto rilevato nell'esecuzione del test di tenuta delle tubazioni a richiesta del TCMS (mancata pressurizzazione/perdita di pressurizzazione)			MAJOR	CRITICO	DEGRADATO	ON	1. Proseguimento operazioni

3. Functional Requirements

3.1 System Configuration

3.1.1 Functional Design

3.1.1.1 System Operating states

The operating states of the system are summarized below:

- AUTOTEST: the system awaits the end of the initial tests and is not available;
- STAND-BY: the system works regularly by monitoring all on-board sensors; Also in the event that the system detects a Major Fault, the system remains in this state and signals the malfunction;
- PRE-ALARM: the system has detected a PRE-ALARM condition;
- ALARM: the system has detected a fire ALARM condition;
- CRITICAL FAULT: The system is unavailable as it is unable to perform any of the alarm detection functions;
- MAINTENANCE: The system goes to this state on command from the external PTE (maintenance PC) through which it is possible to monitor and control every single signal of the system; in this state the System declares itself unavailable as it is not capable to perform any of the alarm detection functions.

3.1.1.2 Passenger/staff areas

Fire detection is performed in passenger saloons, vestibules, toilets, bistrò and staff areas (including staff toilet, driver cabs and electronic rooms). In these zones the fire is detected by using point detectors capable of detecting the presence of smoke and the temperature increase. For this purpose, a proper number of STD is installed in each internal area of the vehicle and their status is acquired by the FDUs.
Each STD includes two sensors (smoke and temperature sensors) in order to guarantee up to 3 functions:

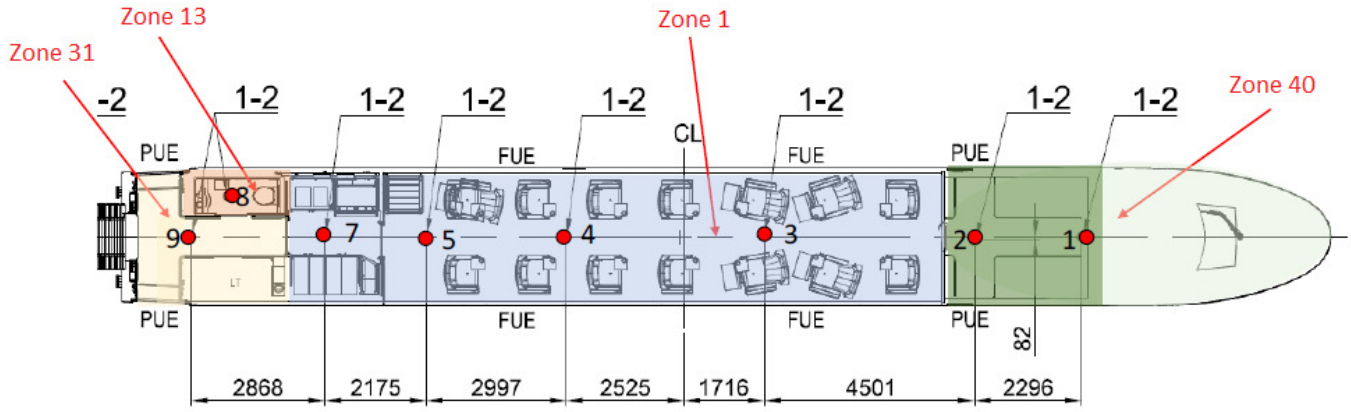
- Smoke detection.
- High temperature detection.
- Quick rise of temperature detection.

The train is divided into Zones which shall be managed separately.
In the following paragraph is shown the distribution of the sensors and the subdivision into areas for all the coaches. Each sensor is associated with its own physical address on the Loop. With these physical addresses the FDU will be able to access for interrogation.

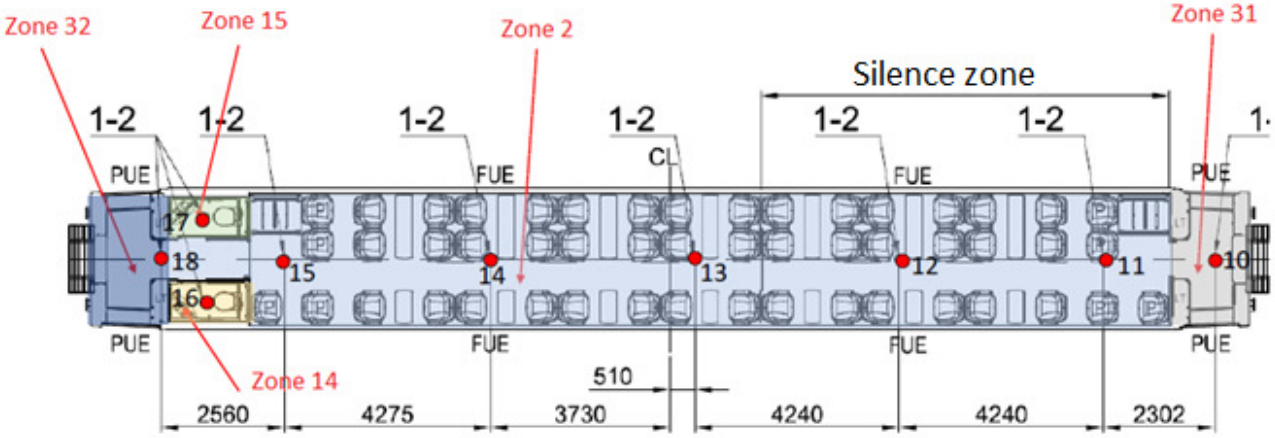
3.1.1.2.1 STD sensors layout

In the following schematics are indicated in red the position of each STD sensor which composed the smoke line along the train.

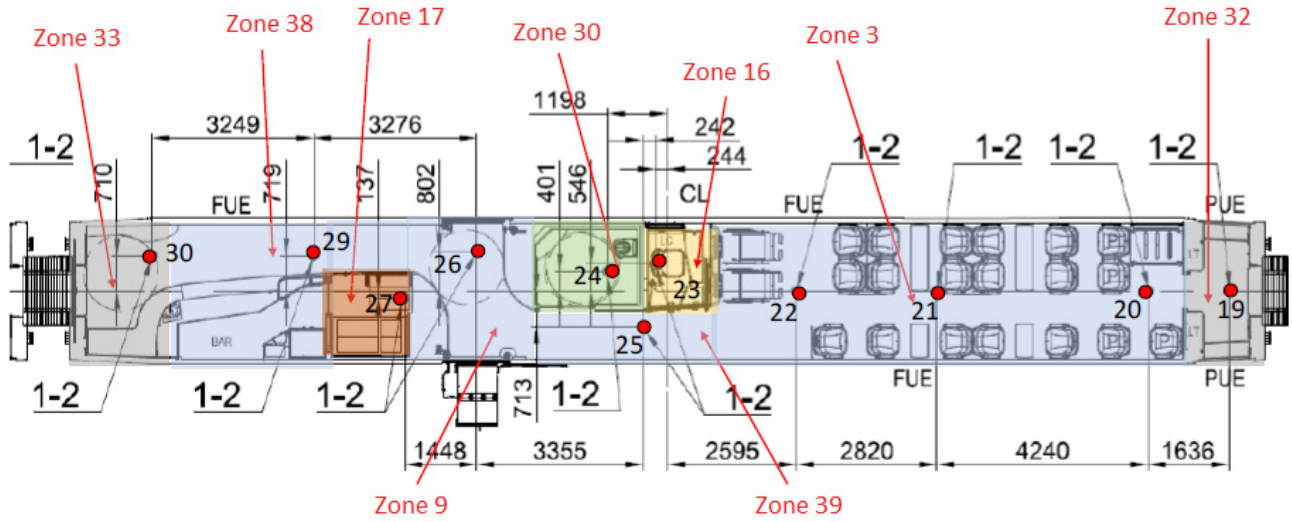
Coach DM1



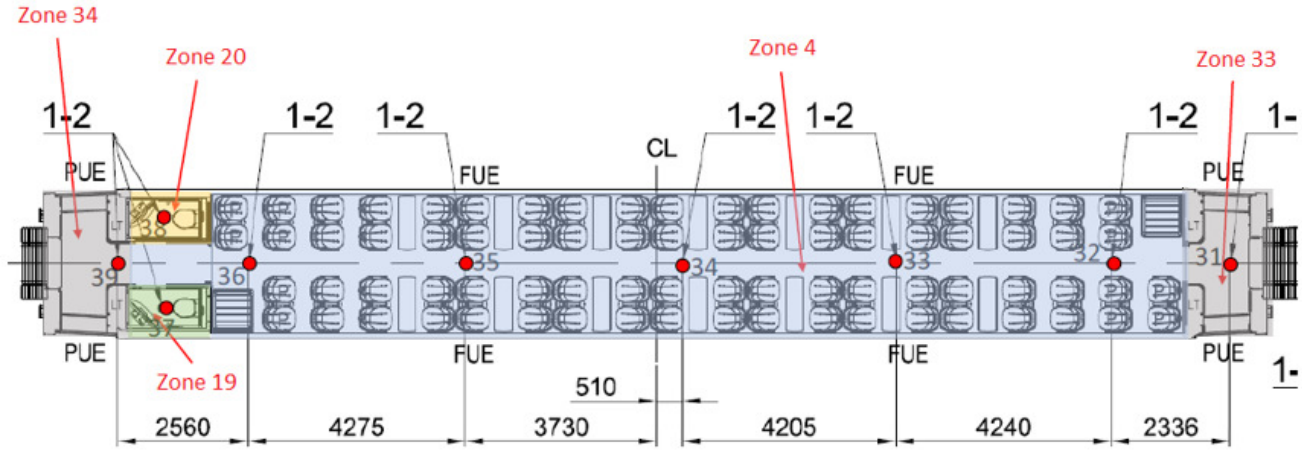
Coach TT2



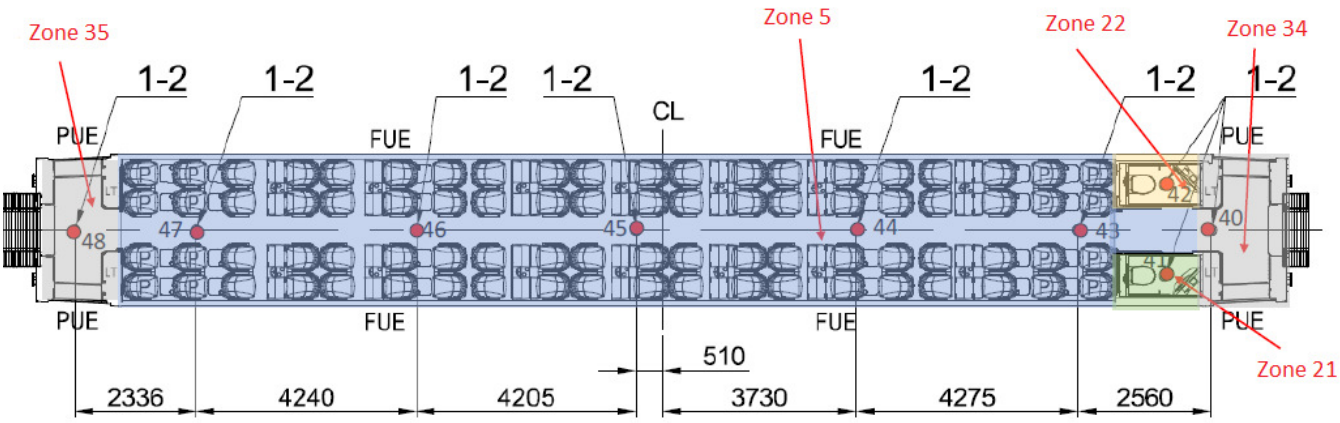
Coach M3



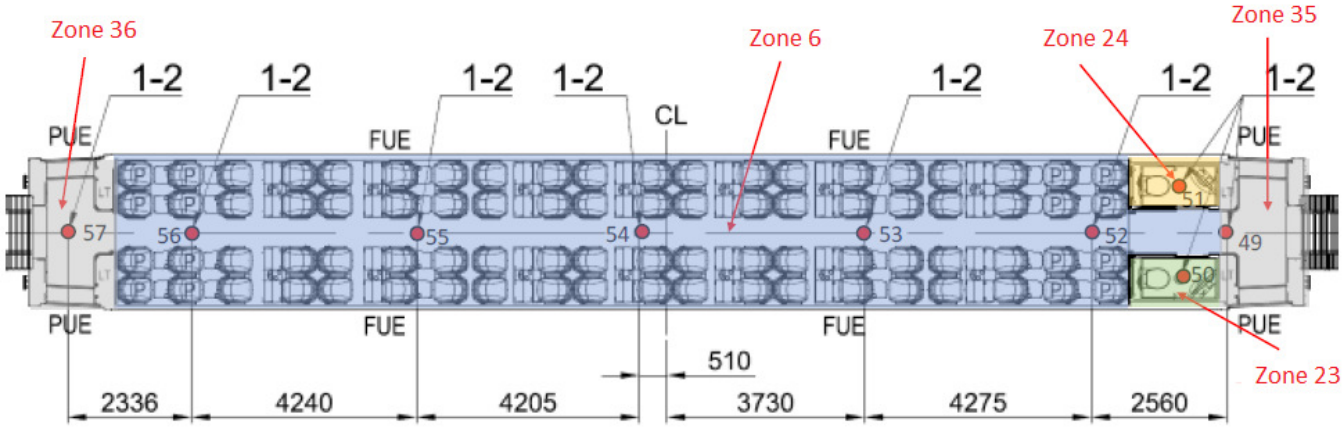
Coach T4



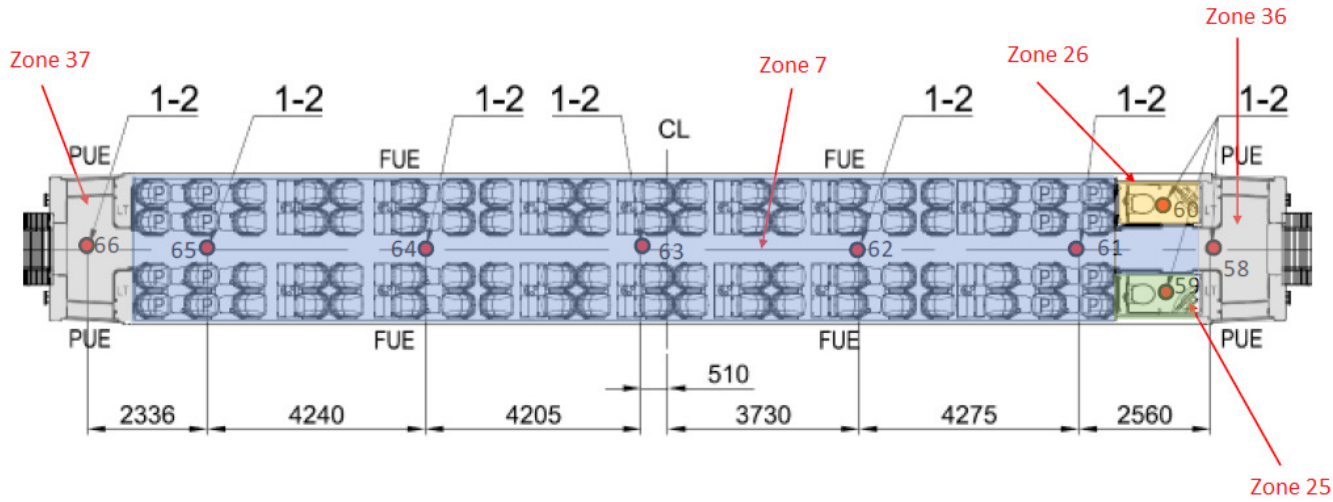
Coach T5



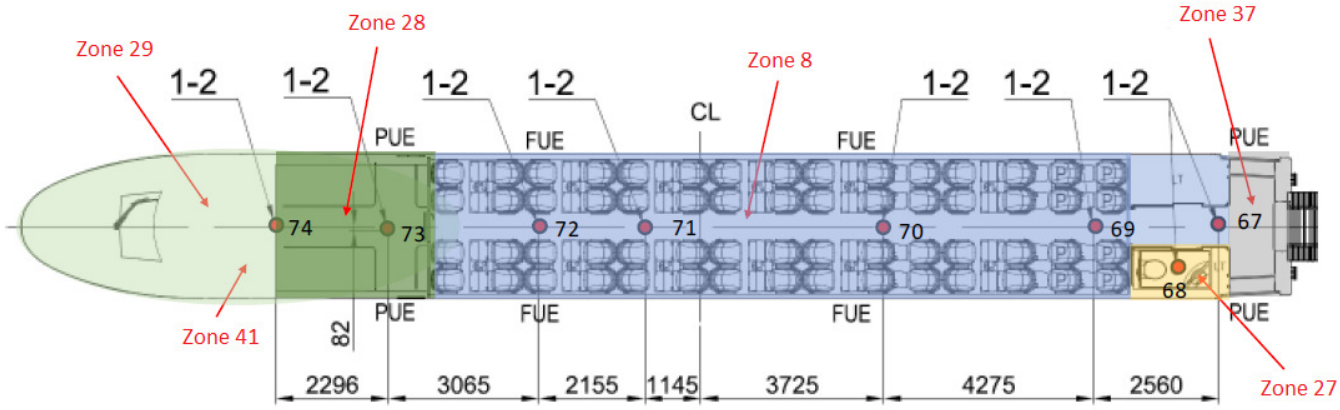
Coach M6



Coach TT7



Coach DM8



For each sensor located in a certain Zone, it is associated a weight related to the exceeding of a smoke and temperature threshold. By adding the weights associated to the sensors belonging to the same zone, the final score that allows the FDU to define a PRE-ALARM or an ALARM status for that zone.

3.1.1.2.2 PRE-ALARM/ALARM managment in passenger/staff area

PRE-ALARM status

The fire PRE-ALARM condition in a Zone is detected if the sum of the weights on the sensors STD associated to this Zone exceeds the score set for the PRE-ALARM condition.
In the PRE-ALARM state, the FPS provides to:

- Communicate the PRE-ALARM status to the TCMS with the indication of the coach involved;
- Indicate the PRE-ALARM status with the LEDs on the FDU;

The Pre-Alarm status remains active for at least 30 seconds. Then the system returns to STAND-BY status once the PRE-ALARM conditions cease.
In the event that the system remains in the PRE-ALARM status due to a faulty sensor, it is possible to exclude this sensor (after 2 minutes) through a dedicated procedure on one of the two FDUs.

ALARM status

The fire ALARM is generated in a Zone when the sum of the weights on the STD sensors exceeds the score set for the ALARM condition.
In ALARM conditions detected in any zone, the Master FDU shall perform the following operations:

1. Direct activation of the ALARM lamp on the driver desk in the local cabin;
2. ALARM lamp activation command on driver desk of DM1 and DM8 via RCL line;
3. ALARM lamp activation command on driver desk in the remote cabin via MCL line;
4. Direct activation of the Alarm-Tx signal on the local coach coupler;
5. Alarm-Tx signal activation command on DM1 and DM8 via RCL line;
6. Activation command of the Alarm-Tx signal on the coupler of the remote coach via MCL line;
7. HVAC deactivation command in ALL the coaches of the train via the RCL line;
8. Communication to the TCMS of the ALARM status with indication of the coach involved;
9. Activation of the LEDs 7 and 8 on the MAIN board of both FDUs.

And, with the exception of the areas relating to the driver's cabs and the electronic cabinets installed in DM1 and DM8:

10. Send the activation command (DV1_FDU) for DV1 valve of WSU;
11. Send the activation command (DV1_RIO) for DV1 valve of WSU via RCL line.

It is NOT foreseen a spontaneous return from the ALARM state to the STAND-BY state but only through operator intervention on FDU according to a dedicated procedure.
If the ALARM conditions persist, it's not possible to restore the STAND-BY state. After 2 minutes it's possible to disable the sensors in alarm status.
The operator can detect on TCMS both the conditions presence/absence of alarm on the STD sensors and the status of the WSU solenoid valve.

3.1.1.2.3 Alarm signaling in passengers/staff areas

As a result of a fire ALARM condition detected in the passenger areas, the FPS system shall:

- Activate the fire ALARM lamp present in each Driver Cab (ALARM);
- Turn off the HVACs of ALL the coaches of the train;
- Rise the alarm to the coupled train (ALARM TX).

3.1.1.3 Technical areas

Fire detection is performed through the implementation of Linear Heat Detector (LHD) cables in all the technical areas installed underfloor and on the roof where electrical devices in band IV (as defined in EN 50153) are present.
In the following table are listed such protected technical areas:

No.	Position	Description	Total number of compartments in 8-car	Locations	Detection presence	Extinguisher presence
1	Underfloor	Line Converter (LL)	8	DM1, M3, M6, DM8	YES	YES
2	Underfloor	Motor Converter (4M)	8	DM1, M3, M6, DM8	YES	YES
3	Underfloor	Converter Cooling (CC)	8	DM1, M3, M6, DM8	YES	YES
4	Underfloor	Auxiliary Converter (AE)	8	DM1, M3, M6, DM8	YES	YES
5	Underfloor	Converter Open Area	4	DM1, M3, M6, DM8	YES	NO
6	Underfloor	DC Main Circuit Breaker (MCB)	2	TT2, TT7	YES	YES
7	Underfloor	Distribution Box (DB)	2	TT2, TT7	YES	YES
8	Underfloor	Trasformer Cooler	2	TT2, TT7	YES	NO
9	Underfloor	Trasformer Control Box	2	TT2, TT7	YES	YES
10	Underfloor	Charging Box	2	TT2, TT7	YES	YES
11	Roof	HV Box	2	T4, T5	YES	YES

Each LHD cable is equipped with End Of Line (EOL) resistor and the FPS system evaluates continuously the impedance of the line.

LHD cables are able to create a short-circuit when the temperature exceeds a certain threshold, thus communicating the presence of a fire in the monitored area.
The FPS system should verify the status of the LHD heat sensitive cable. In particular it shall be able to distinguish the following conditions:

- Open circuit (Fault condition);
- Short-circuit (Fire condition);
- Line connected to the EOL resistor (Safe condition).

If a LHD cable is detected as open circuit, FPS system rises a FAULT alarm.
If a LHD cable is detected as short-circuit, FPS system declares a fire ALARM status in the relevant technical area.

The status and diagnostics of each LHD cable is cyclically detected and communicated to the FDU by pertinent RIO control unit with a cadence of about 3 seconds.

In case of fire ALARM, the activation of the aerosol cartridge is controlled by the FDU Master with a specific timing on relation to the type of technical compartment under protection. In particular, a delay time is implemented between power cuts (commanded immediately) and the associated aerosol activation command.

When the system detects an ALARM condition it continues to monitor the other zones along the vehicle and if in another area detects alarm's condition it will activate the mitigations related to that area.

3.1.1.3.1 ALARM managment in technical areas

In the case of technical areas, the Pre-Alarm condition is not foreseen.

If an ALARM is detected in any technical compartment, the Master FDU provides for perform the following operations:

1. Direct activation of the ALARM lamp on driver desk in the local cabin;
2. Activation of the ALARM lamp on driver desk on DM1 and DM8 via RCL line;
3. Activation of the ALARM lamp on driver desk in the remote cabin via MCL line;
4. Direct activation of the Alarm-Tx signal on coupler of the local coach;
5. Activation of Alarm-Tx signal on DM1 and DM8 via RCL line;
6. Activation of Alarm-Tx signal on the coupler of the remote coach via MCL line;
7. HVAC de-activation in the coach involved with the alarm via the RCL line;
8. Power cut command in the coach involved with the Alarm via RCL line;
9. Aerosol activation command associated to the technical compartment involved after a programmed delay time starting from the power cut;
10. Communication to the TCMS of the ALARM status with indication of the coach(es) involved;
11. LEDs 7 and 8 switched-on in both FDUs;

It is not foreseen a spontaneous return from the ALARM status to the STAND-BY status, but only through operator intervention according to the dedicated procedure.

If the alarm conditions persist, the operator intervention will not have any effect.
The operator can detect on video, through the TCMS, the presence/absence of alarm conditions on LHD cables.

3.1.1.3.2 Alarm signaling in technical areas

As a result of a fire alarm condition detected in the technical areas, the FPS system shall:

- Activate the fire alarm lamp present in each Driver Cab (ALARM);
- Turn off the HVACs of the coach where the fire alarm has been detected;
- Command the Power Shutdown of the coach in alarm;
- Rise the alarm to the coupled train (ALARM TX).

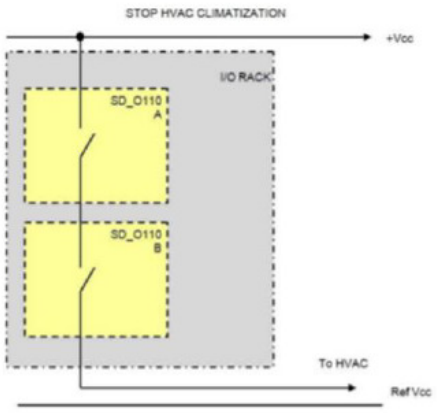
3.1.1.4 Alarm signaling from coupled train

If the system receives the alarm signal from the coupled train, it shall go to the ALARM status and proceed to switch-on the ALARM lamp on the driver desk.
If the FPS is in ALARM status (from the coupled train) detects that no alarms are present anymore from the coupled train, it shall exit from the ALARM status spontaneously.

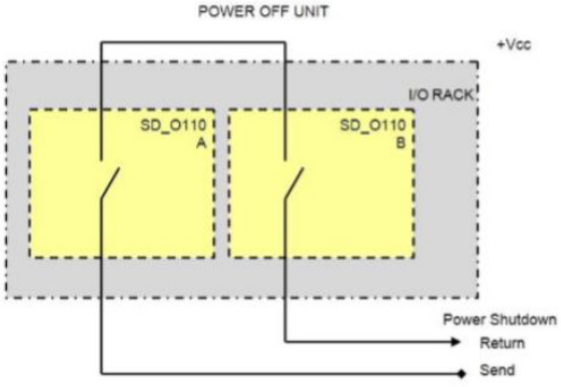
3.1.1.5 Safety hard-line redundancy

The safety hard-lines managed by the RIOs as described above are made with contacts normally open (NO) voltage free.
Redundancy is achieved via a series or parallel connection depending on which negative or positive logic respectively is used.
The safety lines are used to drive the following devices:

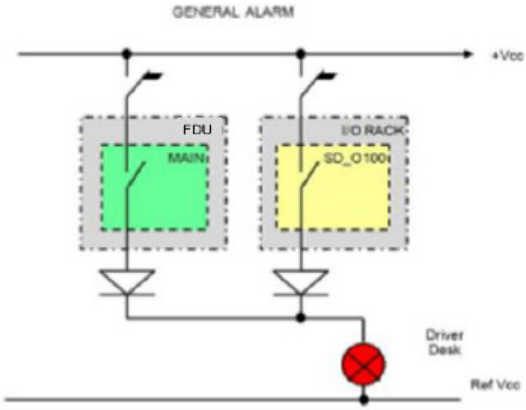
- *HVAC de-activation (negative logic)*



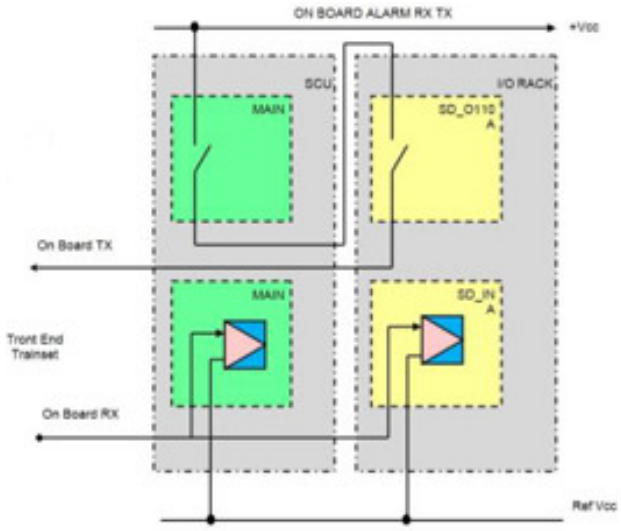
- *Power Shut-down (negative logic)*



- *ALARM signal for lamp on driver desk (positive logic)*



- *ALARM signal for coupled unit (negative logic)*



In the case of contacts managed by the same RIO, different cards are used for minimize common mode risks. The FAULT line does not provide for redundancy, the lamp is controlled only by the RIOs in DM1 and DM8.

3.1.1.6 Fault lines

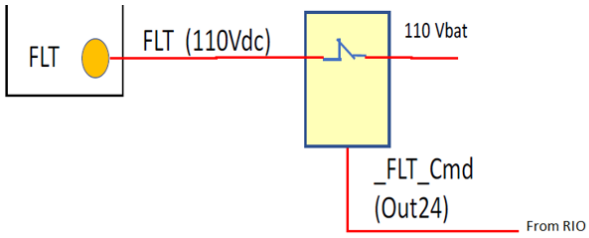
The management of the FAULT lamp on driver desk of each cabin will be managed by remote RIO devices present in the DM1 and DM8 through a 24 Vdc channel. This signal will drive a 24Vdc coil of an external contactor (in HR scope of supply) and which will switch on the "Fault" lamp on the bench and transmission of the "Fault" status to the coupled train.

3.1.1.6.1 FAULT Lamp

The "FAULT" lamp will be driven by an external relay (NC contact):

- 110Vdc → FAULT; 0Vdc → STAND-BY;

The coil of the relay it will be controlled by a 24V output signal available on RIO in the coach DM1/DM8. In absence of command, the FAULT lamp is switched-on.



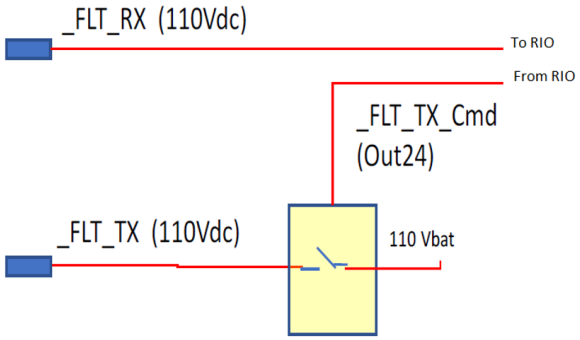
3.1.1.6.2 FAULT from/to coupled train

An external relay (NO contact) in HR scope of supply will used to transmit the “FAULT” status to the coupled train:

- +110Vdc → STANDBY; 0Vdc → FAULT;

The relay coil will be controlled by a 24V output signal available on RIO in the coach DM1/DM8.

The "FAULT" signal coming from the coupled train is acquired in reverse logic from an input channel of the RIOs present in the coaches DM1, DM8 (FAULT_RX).
If there is no coupled train on the header connector, a voltage level of 110Vdc shall be supplied to the "FAULT_RX" input.



3.1.1.6.3 General managment

The two signals (“FAULT” and “FAULT_TX”) are forced to 24V by the RIOs of DM1 and DM8 from the FDU Master in "STAND-BY" conditions. In case of a FAULT, the FDU Master will command the release of the two outputs.
In case the RIO loses the contact with the master FDU, after 10 seconds RIO independently releases the two outputs in order to switch-on the "FAULT" lamp and report the "FAULT" status to the coupled train.

Based on the philosophy indicated in the previous paragraph, the management of the FAULT lines is summarized as indicated in the following table:

Stato RIO	_FLT_cmd	_FLT_TX_Cmd	_FLT_TX	_FLT_RX	_FLT
OFF	0V	0V	0V	0V	110V (On)
Stand-By	24V	24V	110V	110V	0V (Off)
Major/Critical Fault	0V	0V	0V	110V	110V (On)
Fault Rx Treno accoppiato	0V	24V	110V	0V	110V (On)

3.1.1.7 Function Design

Fire detection is performed in passenger saloons, vestibules, toilets, bistrò and staff areas (including staff toilet, driver cabs and electronic rooms). In these zones the fire is detected by using point detectors capable of detecting the presence of smoke and the temperature increase.
Moreover fire detection is performed in all HV and MV cubicles installed underfloor and on the roof (see table at section 3.1.1.3). In these zones the fire is detected by using linear heat cables (LHD).

3.1.1.8 Normal Condition

The STD detectors installed in passenger saloons, vestibules, toilets, bistrò and staff areas provide continuously to the control units an analogical measure of the smoke and temperature level of the supervised area. In case the level of smoke and/or temperature is higher than a pre definite threshold (based on the philosophy explained in par. 3.1.1.2.2) the FPS gives a PRE-ALARM condition followed by the ALARM condition, if reached.
Each passenger/staff area is covered by more than one detector, except for the cabins and toilets where only one detector is installed.
The LHD detectors installed in the cubicles under the vehicle floor or on the roof provide continuously to the control units a digital information regarding their status. If the temperature is higher than the set level, the detector closes a contact and the FPS gives the ALARM status. Each LHD provides and end of line resistor.
In each cubicle only one detector is installed.

3.1.1.9 Failure Cases

a) Loss of function of a smoke/temperature detector in passenger/staff areas.

b) Loss of function of a Line Heat Detector (LHD) in technical areas.

3.1.1.10 Failure Detection

a) Failure condition transmitted to TCMS. In a Zone monitored by only one STD sensor (cabins, toilettes), FPS proceeds to rise a fault message to TCMS via TRDP protocol as ‘‘SYSTEM DEGRADATED’’ in according to TCMS Event report and the FAULT lamp is switched-on in driver desk.

In a Zone monitored by two or more STD sensors, FPS proceeds to rise a ‘‘DIAGNOSTIC’’ Fault to TCMS via TRDP protocol.

b) Failure condition transmitted to TCMS.

3.1.1.11 Failure Handling

Considering that smoke or temperature part of the detector will probably continue to work and that in the passenger area the function is realized also by the adjacent detectors, a redundancy is lost but the function is not endangered. A failure of the detector in a cabin or in a toilet implies that the function is endangered and the driver shall act according to the operational manual.

Temperature detectors in technical areas are not redundant, therefore the function is endangered and the driver shall act according to the operational manual.

3.1.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.96
Description	Smoke/fire shall be detected in toilets.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.67	One smoke/temperature detector shall be foreseen in each toilet.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.133	These detectors shall be connected to the control units through the passenger areas detectors line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_0y.0z.Zefiro-Europe.CONCEPT.HR01	In the event of a PRE-ALARM in areas monitored by only one STD sensor, the FPS system shall communicate the PRE-ALARM status to the TCMS with the indication of the coach.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR52	The PRE-ALARM state remains active for at least 30 seconds. Subsequently the FPS System returns to the STAND-BY state once the PRE-ALARM conditions are ceased.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR53	In the event that the System remains in the PRE-ALARM state (probably due to a failure on the sensor), after 2 minutes, it is possible to exclude this sensor through the disabling procedure.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR57	The FPS shall foreseen the following disabling procedure: - Keep pressed the “Disable” button until LED 3 starts flashing quickly; - Keep pressed the “Reset” button until LED 3 stops flashing quickly.	New Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_0y.0z.Zefiro-Europe.CONCEPT.HR58	The loss of one STD sensor in area is monitored by one or two STD sensor, the FPS system will raise a MAJOR FAULT message to the TCMS via TRDP and and shall activate to +Vbatt the output of the "Fault" lamp on driver's desk.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR59	TCMS shall store an event (SYSTEM DEGRADED) in according to TCMS Event Report in case of FAULT of a STD sensor in area monitored by one or two STD sensors.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR54	In case of fire PRE-ALARM in a toilet, the TCMS sends a command via ethernet to the toilet’s control unit in order to activate the alarm lamp and the buzzer installed in the toilet.	New Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR60	In case of fire PRE-ALARM in a toilet, the TCMS shall command the HVAC unit of the related coach to work in Full Recirculation Mode (HVAC_Full_Recirculation_Mode).	New Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR02	In case of fire ALARM in a toilet, the TCMS sends a command via ethernet to the toilet’s control unit in order to activate the alarm lamp and the buzzer installed in the toilet.	New Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR55	After a PRE-ALARM condition in a toilet, once the FPS System returns to the STAND-BY state, the TCMS shall reset the command via ethernet to the toilet’s control unit in order to de-activate the alarm lamp and the buzzer installed in the toilet.	New Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR09	In the event of a PRE-ALARM in areas monitored by more than one STD sensor, if the FPS remains in the PRE-ALARM state due to a faulty sensor, after 2 minutes, it is possible to exclude this sensor through the disabling procedure.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR03	The only one condition of smoke detection recognized by the STD sensor in toilet shall not cause the ALARM state for the toilet.	New Requirement	4S_06.05.-.Fire Protection Systems	EVENT	N/A

ID	2F_06.04.Zefiro-Europe.TRS.97
Description	Smoke/fire shall be detected in the driver cabs.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.68	One smoke/temperature detector shall be foreseen in both driver cabs.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.134	These detectors shall be connected to the control units through the passenger areas detectors line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.98
Description	Smoke/fire shall be detected in passenger saloons, vestibules
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.69	A suitable number of smoke/temperature detectors shall be foreseen in passenger saloons and vestibules.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.135	These detectors shall be connected to the control units through the passenger areas detectors line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.101
Description	Smoke/fire shall be detected in technical areas where MV or HV devices are installed.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.70	Suitable temperature detectors shall be foreseen to detect possible fire arising in technical boxes containing MV and HV components.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.136	All thermal detectors for technical areas shall communicate to the control units their status.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.106
Description	Smoke/fire shall be detected in electronic room.
Safety level	SIL 2

Electronic room is intended the area between the driver cab and the passenger saloon.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.71	One smoke/temperature detector shall be foreseen in each electronic room.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.139	These detectors shall be connected to the control units through the passenger areas detectors line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

3.2 Supervision of the fire detection system

3.2.1 Function Design

The fire protection system is controlled and commanded by two control units (FDU) installed in car DM1 and car DM8 and working in Master-Slave configuration. The initial state of “master” or “slave” condition is set through a pre-definite cabling on the interface connector that will set Master in car DM1 and slave in car DM8. Then the two control units will alternate periodically their mastership.

The two control units share two independent lines and exchange data on an additional independent serial line (backbone bus).

The first line connects all the detectors of the passenger areas while the second line connects a series of I/O modules. They have the function of collecting alarms from thermal detectors installed in technical areas, commanding consequently the aerosol cartridges and managing the car-based interlocks of HVAC and power disconnection.
In case of line interruption, both units will become Master in the sense that each control unit controls the devices respectively before and after the cable interruption.
In case of short circuit on the line, the interested piece of electric cable will be isolated by the isolators equipping every device on the line, and again both units will become Master, controlling all the devices respectively before and after the isolated cable in short circuit.
In both cases the control units will exchange all relevant data and will align through the independent and dedicated serial line so that the communication with TCMS is done only via the Master control unit.

3.2.1.1 Fault line managment

If the system detects a MAJOR Fault condition, this condition shall be reported to the TCMS as ‘‘System DEGRADATED/OUT OF ORDER’’ message (depending on type of fault) in according to TCMS Event Report and the FAULT lamp on driver desk shall be switched on.
If the system detects a CRITICAL Fault condition, this condition shall be reported to the TCMS as ‘‘System OUT OF ORDER’’ message in according to TCMS Event Report and the FAULT lamp on driver desk shall be switched on.
If the system detects a MAJOR Fault or CRITICAL Fault condition, it must be reported the FAULT status to the coupled train.
If the system receives the FAULT signal from the coupled train, the FAULT lamp on driver desk shall be switched on.
If the system in FAULT status from the coupled train, detects the signal of no fault from the coupled train, shall proceeds to switch off the FAULT lamp on driver desk.

3.2.1.2 Normal Condition

The two control units (FDUs) work in Master/Slave configuration: the Master control unit becomes slave if a fault occurs or after a pre definite period of time. The two control units receive the information about the smoke/temperature levels from the detectors in the passenger areas through the detectors line and send/receive the command/information from/to the LHD cables and aerosol cartriges in the technical areas through the RIO I/O modules line.

3.2.1.3 Failure Cases

a) Loss of function of ~~a~~ FDU control unit in DM1.
b) Loss of function of a FDU control unit in DM8.
c) Interruption/short circuit of a line (RCL or FCL).

d) Loss of communication RS485 for the MASTER FDU

3.2.1.4 Failure Detection

a) Failures are transmitted to TCMS as ‘‘DIAGNOSTIC’’ Fault.This fault will be followed by the change of the FDU’s mastership.

b) Failures are transmitted to TCMS as ‘‘DIAGNOSTIC’’ Fault. This fault will be followed by the change of the FDU’s mastership.

c) Failures are transmitted to TCMS as ‘‘DIAGNOSTIC’’ Fault. The change of the FDU’s mastership is blocked.

d) Failures are transmitted to TCMS as ‘‘DIAGNOSTIC’’ Fault and the FDU’s mastership is blocked.

3.2.1.5 Failure Handling

a) The redundancy is lost and the WSU diagnostic is lost, but FPS is still able to command the extinguish discharge.

b) The redundancy is lost but the complete control of the system is automatically taken by the other control unit so that the complete system functionality is preserved.

c) Both MASTER FDU and SLAVE FDU will be able to interrogate the devices on the line till to the interruption, so no information is lost. Line cabling needs to be checked. The TCMS will continue to receive the diagnostic information ONLY from FDU MASTER.

d) Due to the communication redundacy through the RCL line, no failure handling is foreseen.

3.2.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.99
Description	All smoke/fire detectors shall be supervised by the FPS
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.72	Two control units shall be foreseen working in Master-Slave configuration.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.146	Control units will be identified by a vehicle cabling (jumper) on the interface connector.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.04.Zefiro-Europe.CONCEPT.147	The Master control unit shall become Slave if a fault occurs or after a predefined operation time.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.465	The commutation between Master and Slave control units shall be managed in a safe way avoiding indefined configurations that can endanger the safety of the system.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.332	Both control units shall monitor the presence of the power supply.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.334	In case the control unit realizes that no power supply is available, a diagnostic message shall be sent to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.333	TCMS shall store an event (Maintenance) in case of power supply is missing on either of the control units.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_06.04.Zefiro-Europe.CONCEPT.148	DELETED	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.149	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.150	TCMS shall monitor the communication status of both Master and Slave control unit.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.151	If TCMS finds that there is no communication with either of the control units, an event (Maintenance) shall be stored.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.152	All smoke/temperature detectors (passenger saloon, vestibules, toilets, driver cabs and electronic rooms) shall be connected to both the control units through a communication line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.153	I/O modules shall be foreseen in order to manage the technical areas of the coach (read the status of thermal detectors, command aerosol cartridges) and the local commands to other systems (HVAC, power supply).	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.154	I/O modules shall be connected to both the control units through a communication line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.336	Each I/O module shall monitor the presence of the power supply.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.337	In case the I/O module realizes that no power supply is available, a diagnostic message shall be sent to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.338	TCMS shall store an event (Maintenance) in case of power supply is missing on an I/O module.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.155	The two control units shall share also a communication (RS485) line as a back up.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.45
Description	Failure in the communication lines shall be detected.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.73	The control units shall be able to detect a failure in the detectors line as well as in the I/O modules line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.156	A single point failure in the line shall not affect the communication between either the detectors and the control units or the I/O modules and the control units.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.157	Failures in the line shall be transmitted to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.158	TCMS shall store an event (Maintenance) in case of a failure in the lines.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.100
Description	A failure on a single detector shall not affect the rest of the fire detection system. A single point of failure of the communication line of fire detectors shall not cause the function to fail.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.74	A failure on a smoke/temperature detector STD shall be detected.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

2F_06.04.Zefiro-Europe.CONCEPT.159	A failure on a single detector in passenger areas shall not affect the rest of the fire detection system.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.160	A failure on a LHD shall be detected. A failure on a LHD cable in a technical area shall NOT affect the fire detection capability of the FPS in the other areas.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.104
Description	The fire detection system shall operate in all train modes that have power supply available.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.75	The control units of the fire detection system shall be powered in all train modes that have power supply available.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.3 Driver interface

3.3.1 Smoke/heat alarm

The fire detection system of the passenger compartment is able to recognize a PRE-ALARM status (based on the philosophy explained in par. 3.1.1.2), i.e. a suspicious status where a level of smoke/temperature higher than the nominal is detected. As soon as the level of smoke/temperature exceeds the alarm level the system will get to ALARM status.
The thermal detectors installed in the technical areas are only able to detect the ALARM status.
In case of fire PRE-ALARM status, the system will inform TCMS about which coach is in PRE-ALARM status. TCMS according the received information will command the HVAC unit of the coach where the PRE-ALARM was recognize to work in Full Recirculation Mode (HVAC_Full_Recirculation_Mode) and command the PIS to activate a tone in order to advise the on-board staff.
In case of fire ALARM status is detected in the passenger/staff areas of a coach, following actions are performed:

1. TCMS is informed. A suitable ALARM lamp is switched on and the related buzzer is activated on all driver desks;
2. Both HVAC units of ALL the coaches of the train is commanded to work in Fire Mode, where ventilation is shut-off and external shutters are closed (HVAC_Fire_Mode);
3. If the fire has been detected in the passenger areas, power is interrupted to the equipments positioned inside that coach which can cause electrical shock in case of water mist discharge or damages to components (realized by TCMS);
4. The TCMS will command the PIS to activate a tone in order to advise the on-board staff.

In case of fire ALARM status is detected in any technical areas of a coach, following actions are performed:

1. TCMS is informed. A suitable ALARM lamp is switched on and the related buzzer is activated on all driver desks;
2. Power is shut down in the car where the fire event has been detected in order to NOT feed the fire.

3.3.1.1 Function Design

In order to perform the above described action 1. the Master control unit is alarmed via the information received by the detectors line (passenger areas) or the I/O modules line (technical areas) and activates the HW Fire General Alarm signal (the signal is generated redundantly by the Master control unit and by one I/O module). This signal will light the ALARM lamp and activate the buzzer on the driver desk of the coach where the control unit is mounted. Moreover the Master control unit will communicate the alarm via Ethernet to TCMS which generates an Event on the IDU. The Slave control unit will also activate the HW Fire General Alarm in order to light the ALARM lamp and activate the buzzer on the driver desk of the coach where it is mounted.

In order to light the FIRE GENERAL ALARM lamp and activate the buzzer also on the driver desks of a possible coupled train, N°2 HW signals are foreseen, the first, Fire Onboard Transmitted generated redundantly, for sending the information that the own train is on fire and the second, Fire Onboard Received, for receiving the

information that the other train is on fire. The two signals are sent over the coupler and the Fire Onboard Transmitted of one train is connected to the Fire Onboard Received of the other train.
In particular, each Fire General Alarm signal is managed by a positive logic redundantly generated by HW circuits (NO contact of a supervised relay) in the control unit and in the relevant I/O module, both powered by different power supplies. In order to check the availability of the FIRE GENERAL ALARM lamp power supply, a periodic check of the two feeding lines is foreseen.
In order to perform the above described action 2., the Master control unit, as soon as alarmed by the information received by the detectors line (passenger areas/staff areas) or the I/O modules line (technical areas), commands the relevant I/O modules to activate the HW signal for commanding the HVAC of ALL the coach of the train to work in Fire Mode (HVAC_Fire_Mode).

3.3.1.1.1 Normal conditions

As soon as the Master FDU detects a fire ALARM, it commands the already described outputs on itself and on the relevant I/O modules in order to switch-on the ALARM lamp and activate the buzzer on the driver desks and perform the coach selective interlocks (HVAC, power shut down). All these actions are performed redundantly by the control unit and one I/O module (ALARM lamp and buzzer activation) or by two I/O modules of the affected coach (coach selective interlocks).

3.3.1.1.2 Failure cases

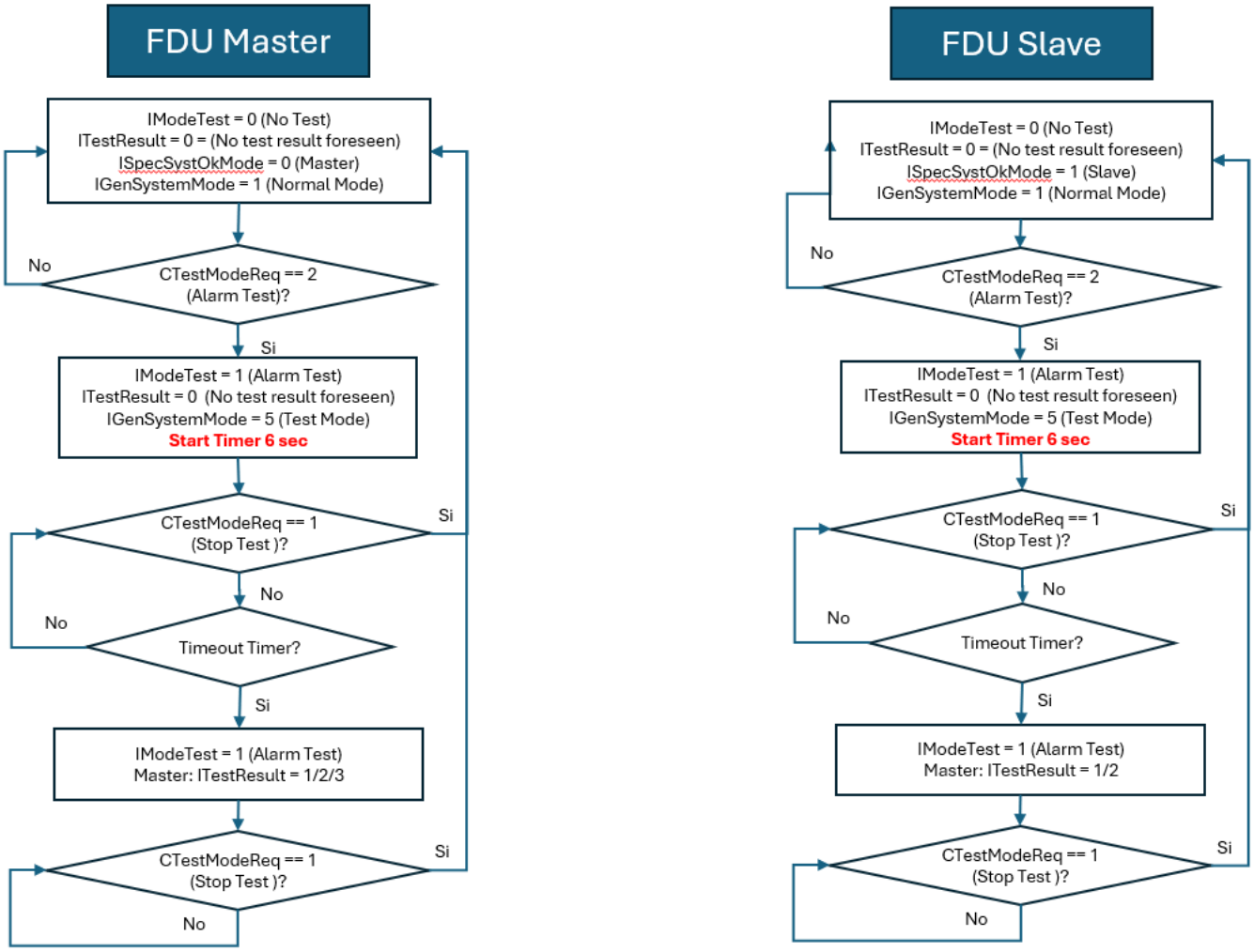
a) Loss of function of ~~a~~ FDU control unit in DM1.
b) Loss of function of ~~a~~ FDU control unit in DM8.
c) Loss of communication of a relevant I/O module.
d) Loss of function of a relevant I/O module.
e) Loss of function of the ALARM lamp on the driver desk.
f) Loss of function of the FAULT lamp on the driver desk.

3.3.1.1.3 Failure detection

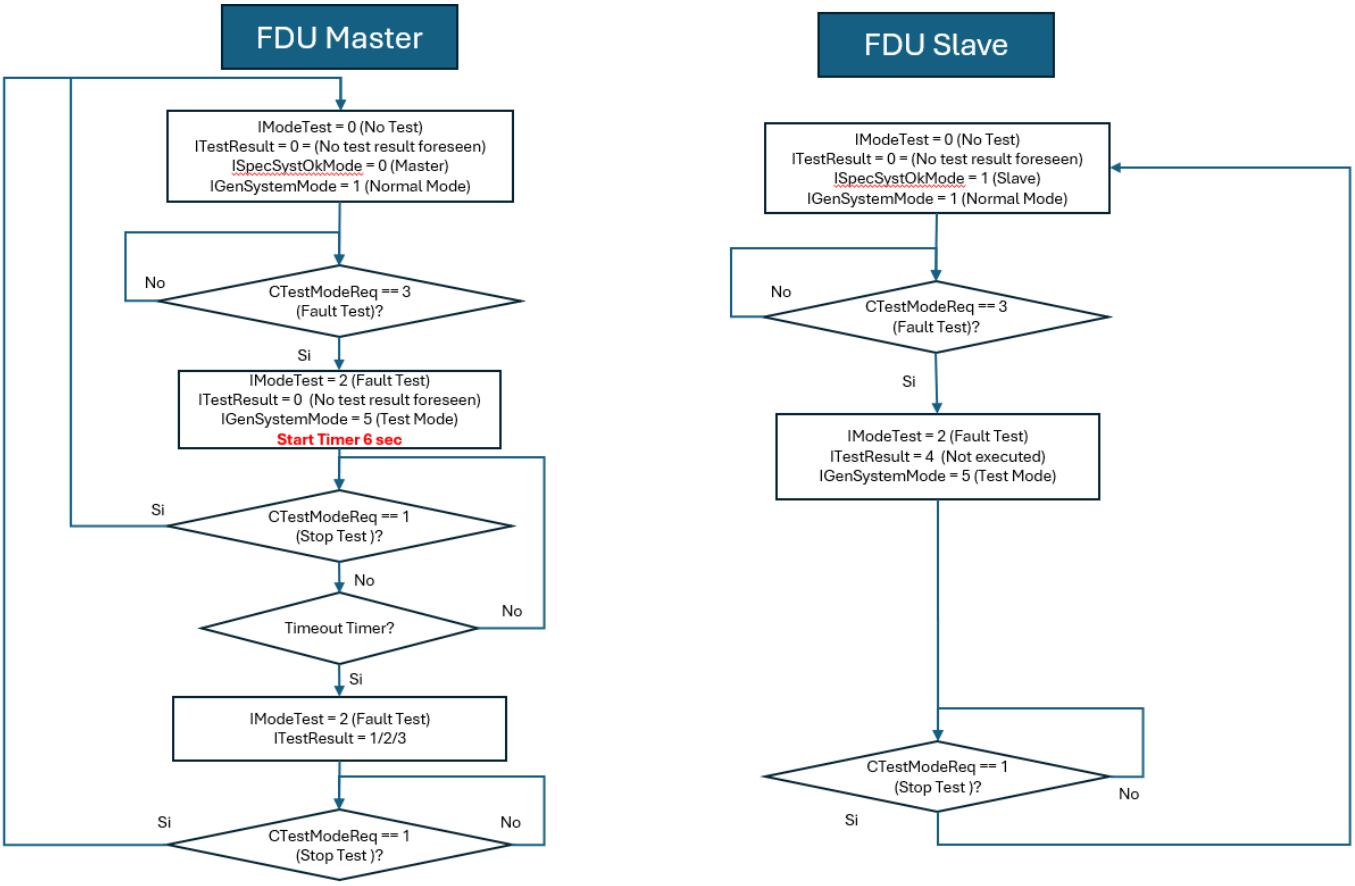
a) See section 3.2.1.4
b) See section 3.2.1.4
c) See section 3.2.1.4
d) The relays realizing the lamps/buzzer activation of and the interlocks are monitored by the control units; a failure is transmitted to TCMS.
e) Failure can only be detected during a suitable test that will be performed periodically (frequency to be defined).
f) Failure can only be detected during a suitable test that will be performed periodically (frequency to be defined).

In the following flow charts are indicated the control logic to be implemented in FPS and the signals interface with TCMS in order to execute the Lamp test for the ALARM lamp and FAULT lamp.

ALARM Lamp Test



FAULT Lamp Test



3.3.1.1.4 Failure handling

a) See section 3.2.1.5

b) See section 3.2.1.5

c) See section 3.2.1.5

d) One redundancy is lost but the function is not endangered. The faulty relay needs to be replaced.

e) The function is lost and the driver shall act according to the operational manual.

f) The function is lost and the driver shall act according to the operational manual.

3.3.1.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.85
Description	When smoke/heat is detected in passanger area and/or in technical area this shall be indicated on the IDU as an event.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.76	TCMS shall store an Event (Alarm) in case of fire detected in any passenger or technical areas indicating which detector and coach is in alarm.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.86
Description	When smoke/heat is detected this shall be indicated to the driver via an optical signal hard wire to the fire detection system. The indication shall be done irrespective of activated cab.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.77	The system shall provide 2 hard wired signals (Fire_General_Alarm) carrying the information that a fire has been detected in any passenger areas or technical areas.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR04	The loss of one STD sensor in a passenger Area monitored by more than two STD sensors, shall NOT affect the fire detection capability. A DIAGNOSTIC fault message shall be available on TCMS in according with TCMS Event Report.	New Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_0y.0z.Zefiro-Europe.CONCEPT.HR61	In case of at least two STD sensors in FAULT in area monitored by more than two STD sensors, the FPS will raise a MAJOR FAULT message to TCMS via TRDP.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR79	In case of at least two STD sensors in FAULT in area monitored by more than two STD sensors, the FPS shall activate to +Vbatt the output of the "Fault" lamp on driver's desk.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR62	TCMS shall store an event (SYSTEM DEGRADED) in according to TCMS Event Report due to a fault of at leat two STD sensors in area monitored by more than two sensors.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR05	The system shall provide 1 hard wired signal (Fault_Alarm) carrying the information to the driver that a fault (MAJOR or CRITICAL) has been detected.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL1
2F_06.04.Zefiro-Europe.CONCEPT.169	The signals _Fire_General_Alarm shall not be affected by the fault of one control unit.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.170	Both signals _Fire_General_Alarm shall be provided in positive logic, i.e. energized as long as fire is detected. The system shall provide 2 hard wired signals (Fire_General_Alarm_Tx) carrying the information that a fire has been detected in any passenger areas or technical areas of the coupled train. Both _Fire_General_Alarm_TX signals must be provided in negative logic, that is, energized until a fire is detected.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.171	A suitable fail safe circuit shall be provided to switch ON a lamp on the driver desk of all cabs as long as the signals _Fire_General_Alarm are energized.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.172	The indication lamp on the driver desks indicating the status of the _Fire_General_Alarm signal shall be possible to test using the Lamp test button	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR06	The indication lamp on the driver desks indicating the status of the Fault Alarm signal shall be possible to test using the Lamp test button	New Requirement	4S_08.07.01.-.Drivers Desk		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR76	The TCMS shall be able to send the command (TC_BO_FDE_X_TstMdReq==3) to test the	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	FAULT lamp after the pressing of the related Lamp test push-button.				
2F_0y.0z.Zefiro-Europe.CONCEPT.HR77	After 20 seconds, the TCMS will send the request to FPS in order to stop the Lamp Test (TC_BO_FDE_X_TstMdReq==1).	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.360	The two internal circuits to the Fire Protection system managing the _Fire_General_Alarm signal generation shall be possible to test separately upon a command received by TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.361	The system shall sent to TCMS the result of the _Fire_General_Alarm test.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR07	The system shall sent to TCMS the result of the Fault Alarm test.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.362	TCMS shall send a command (TC_BO_FDE_X_TstMdReq==2) to the Fire Protection system in order to activate the test of the_Fire_General_Alarm and Fault_Alarm circuits.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.363	TCMS shall store an Event (Warning) if the result of the _Fire_General_Alarm test received from the Master control unit is negative	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR08	TCMS shall store an Event (Warning) if the result of the Fault Alarm test received from the Master control unit is negative	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.173	The information of FIRE ALARM detected (Alarm status) shall be transmitted to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.174	The information of which detector is in Alarm status and which coach it belongs to shall be transmitted to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.175	The system shall provide, on each control unit, 2 hard wired signals (1 input, Fire_Onboard_Received, and 1 output, Fire_Onboard_Transmitted) for receiving/transmitting the information that on the coupled train/on the train there is a fire.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.176	The Fire_Onboard_Transmitted output signal shall be provided in negative logic, i.e. energized as long as no fire is detected in any passenger areas or technical areas.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.177	The Fire_Onboard_Transmitted signal shall not be affected by the fault of one control unit.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.178	The system shall energize the Fire_General_Alarm signal as soon as the Fire_Onboard_Received is low but shall not activate the fire suppression/extinguishing system on the unaffected vehicle.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.179	The 2 hard wired signals for receiving/transmitting the information of Fire_Onboard_Received and Fire_Onboard_Transmitted from one train to the other shall be sent over the coupler.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.04.Zefiro-Europe.CONCEPT.180	Suitable HW logics shall be foreseen to enable the transfer of the Fire_Onboard_Received and Fire_Onboard_Transmitted information when a second train is coupled.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.181	When a suspicious level of smoke/heat is detected in the passenger areas of a coach, the system shall transmit to TCMS the PRE-ALARM status of the system.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.183	TCMS shall store an Event (Warning) in case the information of PRE-ALARM status is received indicating which coach is in Pre-alarm.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.184	TCMS upon the receipt of the fire PRE-ALARM status of a coach shall command the HVAC system of that coach to work in HVAC_Full_Recirculation_Mode.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR10	The FPS shall automatically exit from the PRE-ALARM condition toward the STAND-BY state if the conditions that caused the the PRE-ALARM	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

	are no longer detected (after at least 30 seconds after the PRE-ALARM signal).				
2F_0y.0z.Zefiro-Europe.CONCEPT.HR11	The System must evolve from PRE-ALARM state into the ALARM state if the weight of the STD sensor(s) in alarm reach the preset target for the ALARM state.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR78	In case of fire PRE-ALARM in passenger areas, the TCMS shall command the PIS to activate a tone in ALL the cars of the train in order to advise the on-board staff.	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR79	In case of fire ALARM in passenger areas, the TCMS shall command the PIS to activate a tone in ALL the cars of the train in order to advise the on-board staff.	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.87
Description	When smoke/heat is detected an acoustic alarm shall be started in the activated cab.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.78	TCMS shall set the buzzer in the active driver cab in case of fire detected in any passenger or technical areas.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.185	A buzzer shall be foreseen in order to realize the acoustical warning for the driver in case of fire.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A

ID	2F_06.04.Zefiro-Europe.TRS.88
Description	When smoke/heat is detected the HVAC shall be commanded to “fire mode” via a hard wired signal
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.79	One hardwired output signal for each coach (8 per vehicle) shall be provided to command the HVAC_Fire_mode of ALL the coaches when a fire ALARM has been detected.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.186	Suitable HW logic shall be provided to transmit the command from the 8 outputs of the Fire Protection System to the corresponding HVAC interfaces.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.187	All the 8 hardwired signals shall be energized as long as no fire is detected and shall not be energized as soon as smoke/heat is detected.The HVAC_Fire_mode is kept until the system is reset.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.188	Each HVAC unit shall work in HVAC_Fire_Mode (shut down, disconnect heaters and close external shutters) as soon as the hardwired signal from the Fire Protection System is not energized.	Derived Requirement	4S_13.-.HVAC		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.189	The status of the output command for each HVAC shall be transmitted to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.190	TCMS shall acquire the status of each HVAC_Fire_Mode command provided by the FPS.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.191	The system shall monitor each HVAC_Fire_Mode output command comparing the status and the input of the relay used to implement the function. In case of incoherence the system shall transmit a message of fault to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.192	TCMS shall store an Event (Maintenance) in case of fault of the relay realizing the function of commanding the HVAC systems in HVAC_Fire_Mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.114
Description	When smoke/heat is detected in the passenger areas, power supply shall be interrupted to the equipment that can cause electrical shock.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.370	TCMS upon the information received from the Fire Protection System shall command the power supply shut-off of at least following equipments positioned in the coach where fire is detected: · plugs on the seats (where present); · reading lights on the seats (where present); · PIS monitors; · lighting system (except for emergency functions); · MV electrical devices in the toilets; · MV electrical devices in the bistro area (if M3 is affected).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.110
Description	When smoke/heat is detected in the underfloor technical area power supply shall be interrupted to the technical area.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.120	One hardwired output signal for each coach (8 per vehicle) shall be provided to shut off the power supply to the technical area where fire has been detected.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.213	Suitable HW logic shall be provided to transmit the command from the 8 outputs of the Fire Protection System to the relevant contactors that interlock the power supply to the affected area while running capability shall be preserved.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.04.Zefiro-Europe.CONCEPT.214	All the 8 hardwired signals shall be provided as free voltage contacts managed in negative logic, i.e. energized as long as no fire is detected in any technical area of the coach.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.215	The system shall monitor each output signal comparing the status and the input of the relay used to implement the function. In case of incoherence the system shall transmit a message of fault to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.216	TCMS shall store an Event (Maintenance) in case of fault of the relay realizing the function of shutting off the power supply to the technical areas.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.369	TCMS upon the information received from the Fire Protection System shall command the power supply switch off to the affected area.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3.2 Mute Function

3.3.2.1 Function Design

Due to the fact that the fire alarm is classified as an ALARM event, the mute function is performed by the driver when he acknowledges the event.

3.3.2.1.1 Normal conditions

In normal condition the fire warning buzzer can be muted by a suitable command available to the driver. This is not a safety relevant function and the causes of possible loss of the function are not under the Fire Protection System control.

3.3.2.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.73				
Description	The driver shall have the possibility to mute sound alarm in case of smoke/heat detection.				
Safety level	SIL 0				
Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.04.Zefiro-Europe.CONCEPT.80	The same button used to mute other acoustical warnings in the driver cab can be used to mute also the fire buzzer.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3.3 Reset Function

3.3.3.1 Function Design

A suitable push button is foreseen on each control unit in order to electrically reset the system. The electrical reset of the system involves the reset of the status of the smoke/temperature detectors for passenger areas (if any was activated), the reset of the Fire_General_Alarm and Fire_Onboard_Transmitted signals and of the commands to HVAC, and power disconnection systems.

3.3.3.1.1 Normal conditions

As soon as the push button on one of the two control units is pressed and fire conditions are no longer present, the system will reset the ALARM status.

3.3.3.1.2 Failure cases

a) Fault on the RESET push button.

b) Stuck contact of one of the relays generating the _Fire_General_Alarm management functions.

3.3.3.1.3 Failure detection

a) Electrical continuity of the RESET push button is monitored by an end of line resistor.

b) Relay implementing the function is monitored, TCMS is informed. The ALARM lamps on the driver desks are permanently lit.

3.3.3.1.4 Failure handling

a) The function is lost and the driver shall act according the operational manual.

b) The function is lost and the driver shall act according the operational manual.

3.3.3.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.41
Description	The smoke/heat optical signal shall be switched off when the system is reset.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.81	A push button on both control units shall be foreseen in order to reset the system following an alarm.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.433	Each push button shall be monitored. In case of fault a diagnostic message shall be sent to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.434	TCMS shall store an event (Maintenance) in case of fault of the reset push button.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.219	As soon as the push button is pressed on either of the control units and fire conditions are no longer present, the electrical reset of the system shall be accomplished.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR44	In ALARM status (detected in passenger compartment or technical area) and the operator press the RESET push-button, if the alarm's causes has been ceased, then the FPS shall: - Set to 0V both signals _Fire_General_Alarm; - Set the "AlarmTx 1" and "AlarmTx 2" outputs to +Vbatt - Reset the information to the TCMS via bus regarding the affected area; - Reset the alarm status of the stimulated sensors to the TCMS via bus; - Reset the "HVAC_Fire_mode" output to +Vbatt; - Reset the "activated" status of the command for the "HVAC_Fire_mode" line via bus to the TCMS. - Switch-off the ALARM lamp on driver's desk;	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR45	In ALARM status (detected in passenger compartment or technical area) and the operator press the RESET push-button, if the alarm's causes are still present, then the FPS shall: - Maintain to +Vbatt both signals _Fire_General_Alarm; - Maintain to 0V the "AlarmTx 1" and "AlarmTx 2" outputs; - Maintain active the ALARM information to the TCMS via bus; - Maintain active the ALARM status of the stimulated sensors to the TCMS via bus; - Maintain the "HVAC_Fire_mode" output to 0V of ALL the cars of the train; - Maintain the "activated" status of the command for the "HVAC_Fire_mode" line via bus to the TCMS. - Maintain switched-on the ALARM lamp on driver's desk;	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR56	After a successfully ALARM RESET, the TCMS shall: - Reset on TOD the signalization of fire ALARM status of car involved; - Reset on TOD the fire ALARM status of the STD sensor involved.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.84
Description	It shall not be possible to switch off smoke/heat alarm until the system is reset.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.82	Status of the _Fire_General_Alarm and Fire_Onboard_Transmitted signals shall be reset when the reset push button on either one of the the control unit is pressed and fire conditions are not present.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.435	The signal _Fire_General_Alarm shall not be de activated by any HW or SW circuits.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.3.4 Display detectors status on IDU

3.3.4.1 Function Design

This function provides all the information regarding the status of the complete Fire Protection System and the system detectors to the TCMS in order to be displayed on a suitable IDU page. This is not a safety relevant function.

3.3.4.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.64
Description	The status of the fire detection system shall be indicated on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.479	The system shall transmit to TCMS its functional status using the ICD available variable IGenSystemMode.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.83	The system shall transmit to TCMS its status (Starting, Stand-by, Pre-alarm, Alarm, Suppression, Extinguishing (after delay time), Discharge disabled, Maintenance/Test).	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.221	TCMS shall store the Fire Protection System status (Starting, Stand-by, Pre-alarm, Alarm, Suppression, Extinguishing (after delay time), Discharge disabled, Maintenance/Test) and show it on a suitable IDU page.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.480	TCMS shall store the functional status of the Fire Protection System using the variable IGenSystemMode received by the Master control unit which is identified by the variable ISpecSystOkMode=0.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.481	TCMS shall lit the "Out of order" dedicated icon on the IDU as soon as the Master control unit provides the information that the functional status of the system is set to Out of Order or Test/Maintenance (IGenSystemMode=3 OR 10) or when both main circuit breakers relevant to the control units are tripped	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.482	TCMS shall store an Event (Warning) in case the functional status of the system is set to Out of Order (IGenSystemMode=3).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.483	TCMS shall store an Event (Warning) in case the functional status of the system is set to Test/Maintenance (IGenSystemMode=10).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.484	TCMS shall store an Event (Warning) in case the functional status of the system is set to Degraded (IGenSystemMode=2).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.68
Description	The status of the fire/smoke detectors shall be indicated on the IDU (detecting smoke/heat, detector faulty).
Safety level	SIL 0

The LHD and the aersol cartridges need to be replaced after an alarm.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.84	The system shall transmit to TCMS the status of all detectors installed in the vehicle: · smoke/temperature detectors (Normal, Smoke alarm, Thermal alarm, Fault, Sensor disabled); · linear heat cables (Normal, Alarm, Fault).	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.223	TCMS shall receive the status of all detectors installed in the vehicle: · smoke/temperature detectors (Normal, Smoke alarm, Thermal alarm, Fault, Sensor disabled); · linear heat cables (Normal, Alarm, Fault) and show it on a vehicle synoptic chart on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.224	TCMS shall store an Event (Maintenance) in case the status of a smoke/temperature detector of the passenger areas is set to Disabled or Fault.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.225	TCMS shall store an Event (Maintenance) in case the status of a LHD is set to Alarm or Fault.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.466	The system shall transmit to TCMS the status of all aerosol cartridges installed in the vehicle (Normal, Activated, Fault).	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

2F_06.04.Zefiro-Europe.CONCEPT.467	TCMS shall receive the status of all aerosol cartridges installed in the vehicle (Normal, Activated, Fault) and show it on a vehicle synoptic chart on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.468	TCMS shall store an Event (Maintenance) in case the status of an aerosol cartridge is set to Activated or Fault.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR14	The LHD sensor cables shall be constantly monitored by the FPS through impedance measurement in order to detect the following anomalous conditions: • Open Line • No stable value (Out of Range) • Closed Line (Alam) • No stable channel (Circuit error) If an anomalous conditions is detected, the FPS shall rise a MAJOR fault message to TCMS, indicating the location of the LHD cable.	New Requirement	4S_06.05.-.Fire Protection Systems	~~EVENT~~	SIL 1
2F_0y.0z.Zefiro-Europe.CONCEPT.HR15	TCMS shall store an Event (Maintenance) as ''DEGRADATED'' in according with TCMS Event Report in case of ''Major fault''due to LHD in Fault.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR16	The System shall periodically verify the integrity of the Aerosol by measuring the impedance in order to detect the following conditions: • Open circuit: - Fault of the aerosol (MAJOR fault); - Aerosol explosion; • Short circuit: - Fault of the aerosol (MAJOR fault); • Nominal impedance: - Aerosol in healty; If the System detects an open circuit condition, it shall be able to discriminate, based on a previous activation command, the fault or exploded aerosol condition.	New Requirement	4S_06.05.-.Fire Protection Systems	~~EVENT~~	SIL 1
2F_0y.0z.Zefiro-Europe.CONCEPT.HR18	At startup, the system shall detect any anomalies on STD sensors (dirty sensor) based on exceeding certain thresholds.	New Requirement	4S_06.05.-.Fire Protection Systems	~~EVENT~~	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR42	In case of an ALARM occurs at startup, the FPS recognize it as a fault.	New Requirement	4S_06.05.-.Fire Protection Systems	~~EVENT~~	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR19	At startup, the FPS shall detects any anomalies related to LHD, such as impedance out of tolerance, based on the exceeding of certain thresholds. Furthermore, any short circuit detected at startup is identified as a fault and not as an alarm.	New Requirement	4S_06.05.-.Fire Protection Systems	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR20	In case of ALARM conditions in the Technical Areas, the FPS shall command: - the activation of the ALARM lamp on both DM1 carriages and DM8; - switch-off the HVAC of the car where the fire alarm has been detected; - activation of the Fire on Board signal onBoardTx on both carriages DM1 and DM8; - switch-off the power supply of the car where the fire alarm has been detected.	New Requirement	4S_06.05.-.Fire Protection Systems	EVENT	SIL 2

3.4 Fire Extinguisher

3.4.1 Function Design

3.4.1.1 Passenger/staff areas

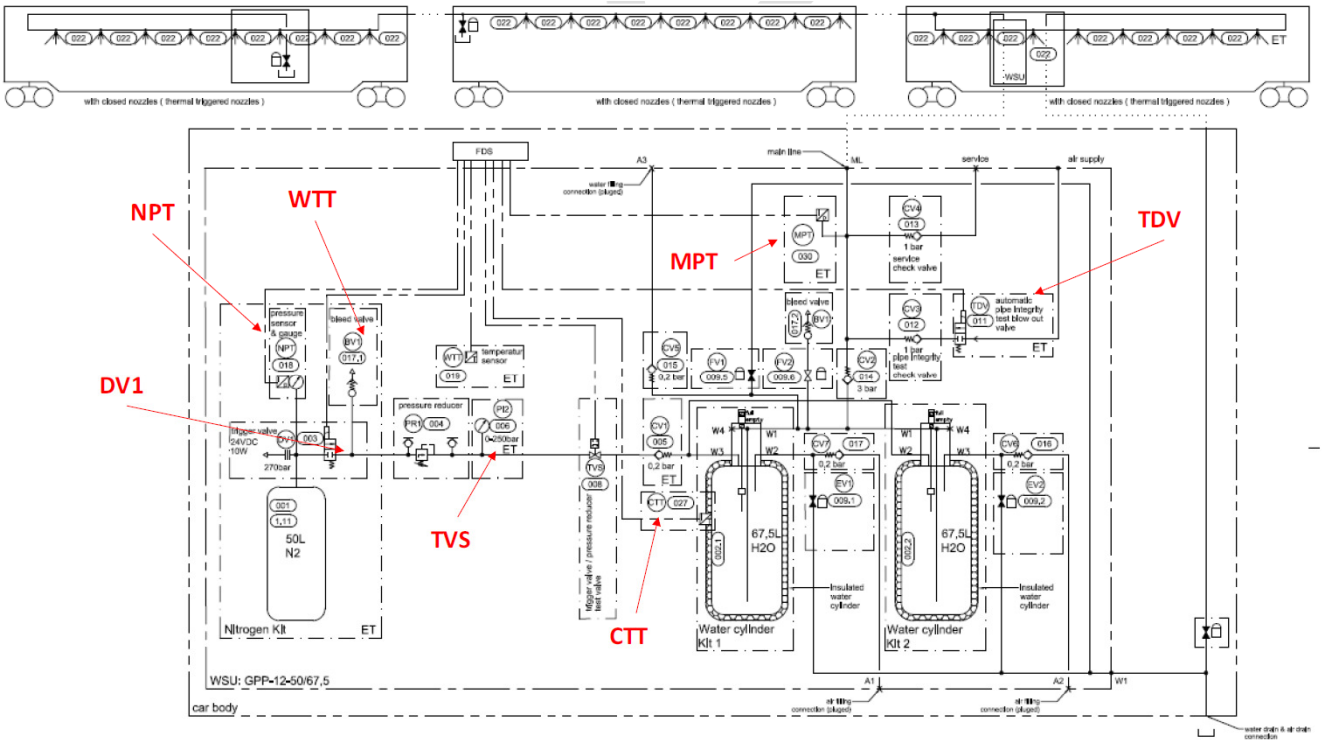
The fire suppression system in passenger/staff areas is based on high pressure water mist technology. Main components of the system are:

- GPP water pakage containing water cylinders (135L)
- Nitrogen cylinder kit (50L)

- Sprinklers (with the related adapters and protection covers)
- Stainless steel piping network to suitable sprinklers for the water mist discharge.

One Water Supply Unit (WSU) is foreseen, positioned in DM1. The piping network is constituted by a main branch connecting the unit along the whole train where special nozzles, of sprinkler type including a thermal sensitive bulb, are connected.

In the event that an alarm condition is detected in any area, the rapid flow valve DV1 placed on Nitrogen cylinder is opened and nitrogen flows at a constant pressure inside the water bottles. The water is then conveyed into the main pipe. In the picture below is shown the WSU hydraulic schematic.



The FPS system shall able to manage the WSU through the following exchanged signals:

Signals from WSU to FDU

- 4-20mA signal from pressure transducer placed on Nitrogen cylinder (**NPT**);
- 4-20mA signal from ambient temperature transducer detected inside the WSU (**WTT**);
- 4-20mA signal from temperature transducer placed on water cylinder (**CTT**);
- 4-20mA signal from pressure transducer inside the main pipe (**MPT**);
- Digital signal in order to monitor the status of the manual valve for the water charge/discharge (**TVS**).

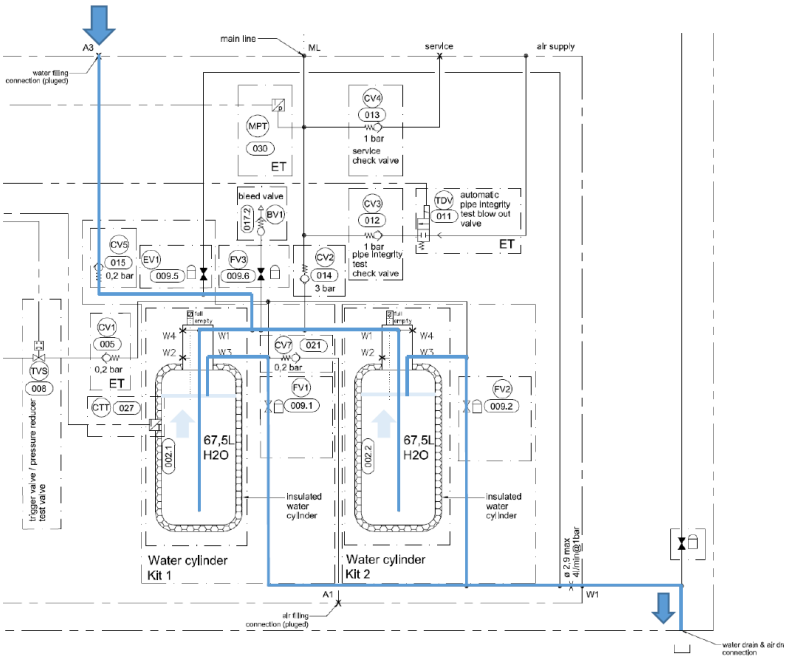
Signals from FDU to WSU

- Command to energize the solenoid valve for water release placed on WSU unit (**DV1**);
- Command for the activation of the compressed air (**TDV**);

3.4.1.1.1 WSU Filling & Draining procedures

Filling

Filling has to be done over to the foreseen filling connection. Clear tap water has to be secured (filtration <= 100 µm). On relation with the following schematic, below is indicated the correct filling procedure.

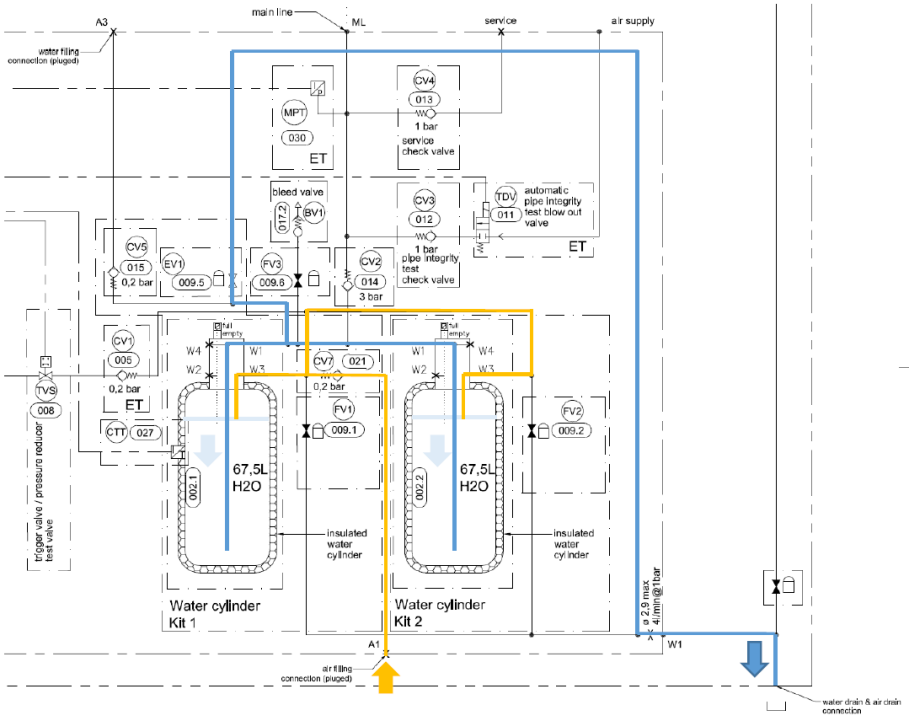


1. Remove the plug on connection A3 and connect the filling hose. Do not start water supply;
2. Connect drain hoses to the out let of the train. To collect outflowing water after reaching filling level, a suitable container has to be used (capacity at least 50 l). Place the container on the floor level;
3. Remove the fixation screw of ball valve FV3 and close it;
4. Remove the fixation screw of ball valve FV1 and FV2 and open them;
5. Start water supply, water flow starts. Maximum allowed pressure of incoming water has to be considered (1 bar);
   During first phase of filling process, air will come out on connection the outlet of the train;

6. Wait until a constant and air bubble free extinguishing agent flows out on connection the out let of the train. Then wait an additional time of 5 minutes;
7. Stop the water supply and disconnect the filling hose from the connection A3;
8. Close the ball valve FV1 and FV2 at the same time;
9. Connect compressed air of 1 bar to the connection A3;
10. Open FV1 and FV2 at the same time and wait until no more extinguishing agent flows the outlet of the train;
11. Stop the compressed air supply and disconnect the compressed air hose from the connection A3;
12. Check the level with the level indicator. Use a resistance (Ohm) meter with measuring tips:
    - If a valve of app. > 200 kΩ is indicated the level is reached (the value depends on the water quality);
    - If the meter shows < 200 kΩ the correct level is not reached. Repeat the process.
13. Open ball valve FV3 and mount the fixation screws on both ball valves;
14. Close the ball valve FV1 and FV2 and mount the fixation screws on both ball valves;
15. Disconnect the drain hose from the outlet of the train;
16. Remount plug on A3;
17. If a pressure over 1 bar occured during the filling process the main line needs to be drained accordingly.

Draining

On relation with the following schematic, below is indicated the correct draining procedure.



1. Connect the drain hose to the outlet of the train. To collect outflowing water after reaching the filling level, a suitable container has to be used (capacity at least size of water cylinder). Place the container on the floor level.
2. Remove the fixation screw of the ball valve FV3 and close it;
3. Remove the fixation screw of the ball valve EV1 and open it;
4. Remove the plug on connection A1 and connect compressed air to connection;
5. Start compressed air supply (1bar is the maximum allowed pressure of the compressed air);
6. Wait until no more water runs out the outlet of the train plus 1 additional minute;
7. Stop compressed air supply;
8. Disconnect compressed air to connection and install the plug on connection A1;
9. Disconnect drain hoses to the outlet of the train;
10. Open ball valve FV3 and mount the fixation screw;

11. Close the ball valve EV1 and mount the fixation screw;
12. If a pressure over 1 bar occurred during the emptying process drain the main line accordingly.

3.4.1.1.2 WSU redundant pre-activation

Pre-activation on the WSU consists of opening of the DV1 solenoid valve which connects the pressurized nitrogen cylinder with the water cylinders.

The WSU valve is piloted separately from the FDU and RIO present in the DM1 box. Therefore, in the event of a fire alarm detection in passenger compartment, the FDU Master (if located in DM1), shall:

- Autonomously command the EV2 valve;
- Command the piloting of valve EV1 trough a command to the RIO of DM1 via RCL line.

If the FDU Master is located in DM8, in the event of a fire alarm detection in the passenger compartment it shall:

- Command the piloting of valve EV2 in DM1 through the MCL and RCL line;
- Command the piloting of valve EV1 trough a command to the RIO of DM1 via RCL line.

3.4.1.1.3 Fire suppression system restore procedure

Below are indicated the restore procedures for the WSU and piping network that the operator shall follow after a water mist discharge:

WSU:

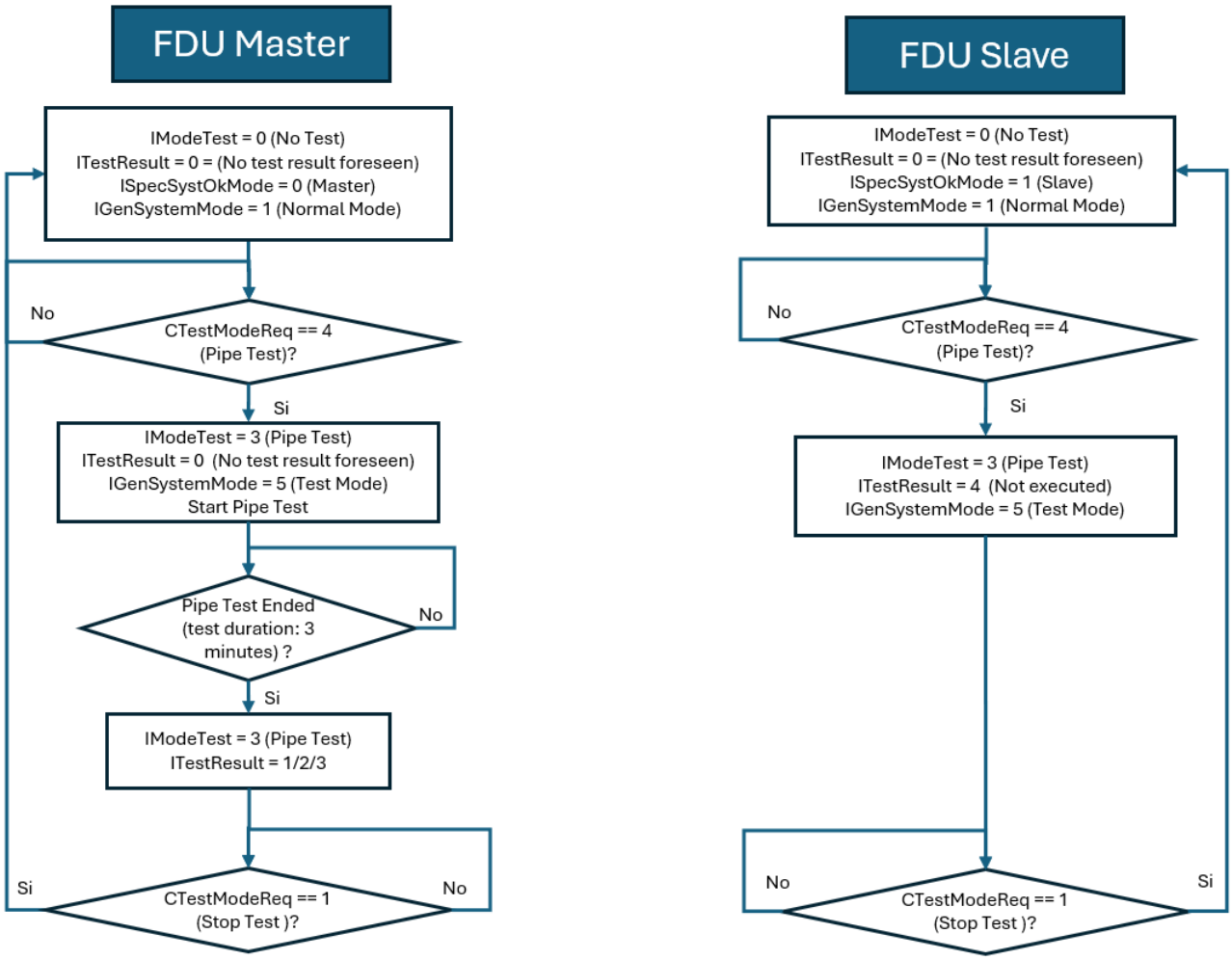
1) Depressurize system on gas and water side;
2) Remove empty nitrogen cylinder;
3) Replace nitrogen cylinder with new or refilled nitrogen cylinder;
4) Execute trigger valve test (or tightness test gas side);
5) Refill extinguishing agent;

PIPING:

1) Change the triggered nozzle heads with new ones;
2) Drain main line from water with compressed air (section by section);
3) Check nozzle heads in the whole train (glas bulb in place);
4) Refill main line with compressed air / wait 30 min and check if a pressure drop occurs.

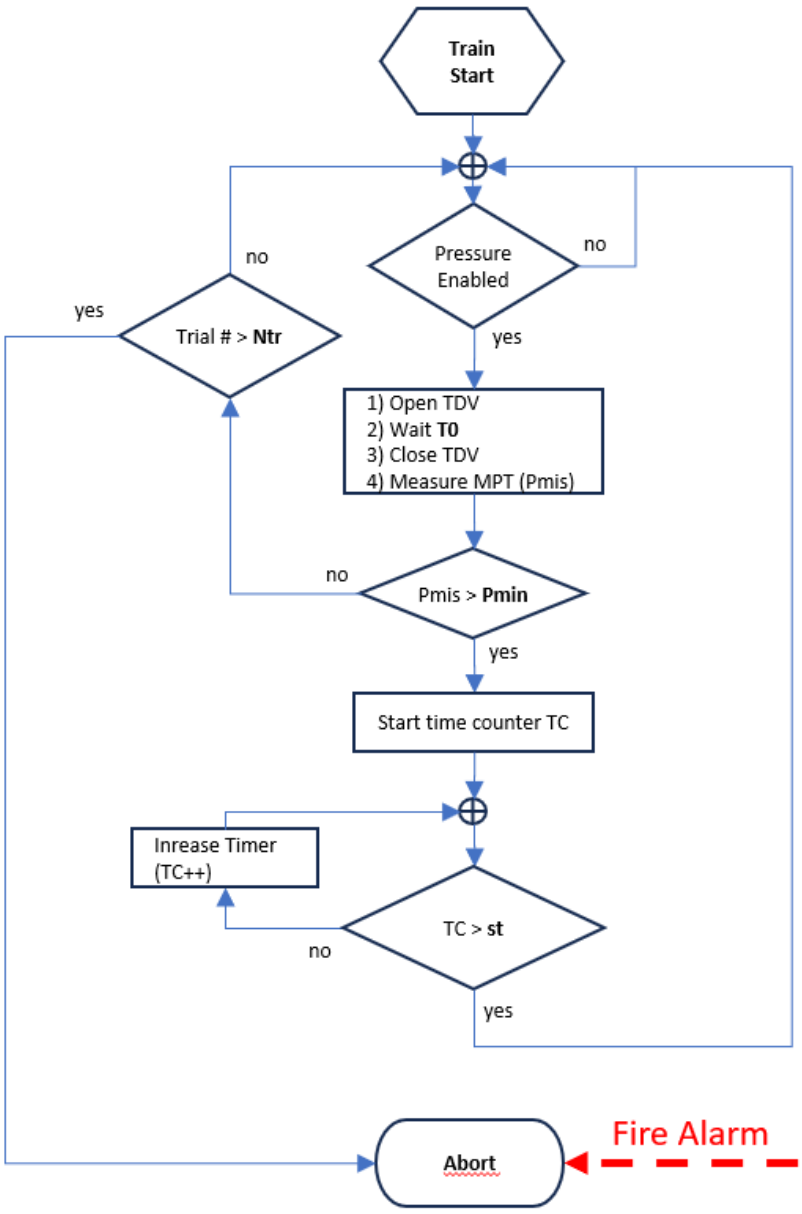
3.4.1.1.4 Piping Integrity Test

In preventive maintenance slot (**I0**), a Piping Integrity Test shall be carried out for the fire suppression conduct. A maintenance page on TOD is foreseen for the FPS for the operator to command the execution of the test, that will be performed automatically by the FDU.
On maintenance page on TOD is implemented a button to enable the Piping Integrity Test. The FDU control unit will receive the command to proceed with a pre-defined algorithm implemented in FDU, by opening the compressed air intake valve (TDV) and reading the value from the pipeline pressure transducer (MPT) over time (typically a few minute).
During the execution of the procedure, the status "TEST ACTIVE" will be indicated on the TOD. Once the procedure has been successfully completed, the status "TEST OK" will be indicated on the TOD if no leakages are detected.
In the event of a test failure, the TOD will indicate the status "FAULT PIPE PRESSURING" and consequently the FAULT lamp on driver's desk will be switched on.
Below is indicated the flow chart of the control logic to be implemented in the FPS in order to execute the Piping Integrity test.



3.4.1.1.5 Compressed air supply during train service

During the train service, the compressed air will be provide within a range from 1 to 4 bar to the piping of the suppression system by opening of the TDV valve. During the train service, the TDV valve will be controlled by TCMS (through the FDU of FPS) based on the flowchart below, which defines the algorithm to be implemented in TCMS software. If the compressed air from main line is not available or the TCMS will recognize that it's not possible guarantee at least 1 bar in the conduct, no fault will be generated and any sealing problems will be managed during the first maintenance available (**I0**).



Legend:

T0 = Minimum time needed for pressure settling inside the piping in absence of significant leakage (supplied by AS) → i.e. 3mins.

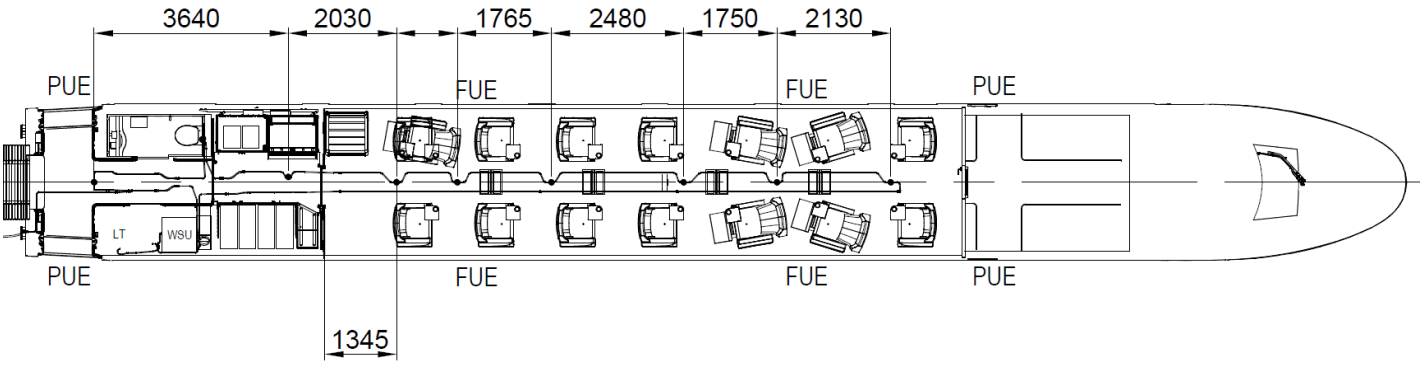
Pmin = Minimum level of nominal pressure required by AS for bulbs correct operation (supplied by AS) → i.e. = 1bar.

st = Sample time (supplied by AS/Hitachi) → To be adjust on field.

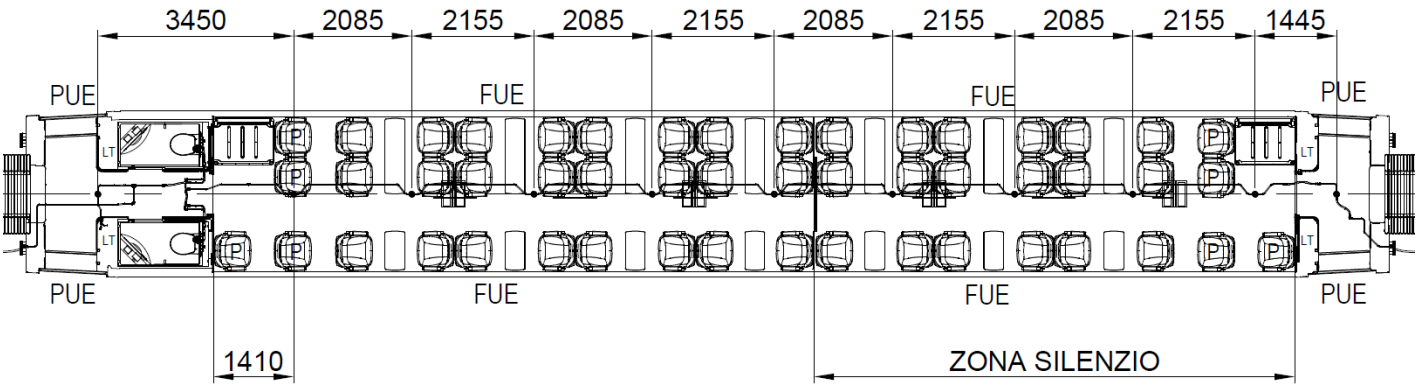
Ntr = Number of trials (supplied by Hitachi) → i.e. = 1 tentative.

3.4.1.1.6 Fire extinguisher system layout

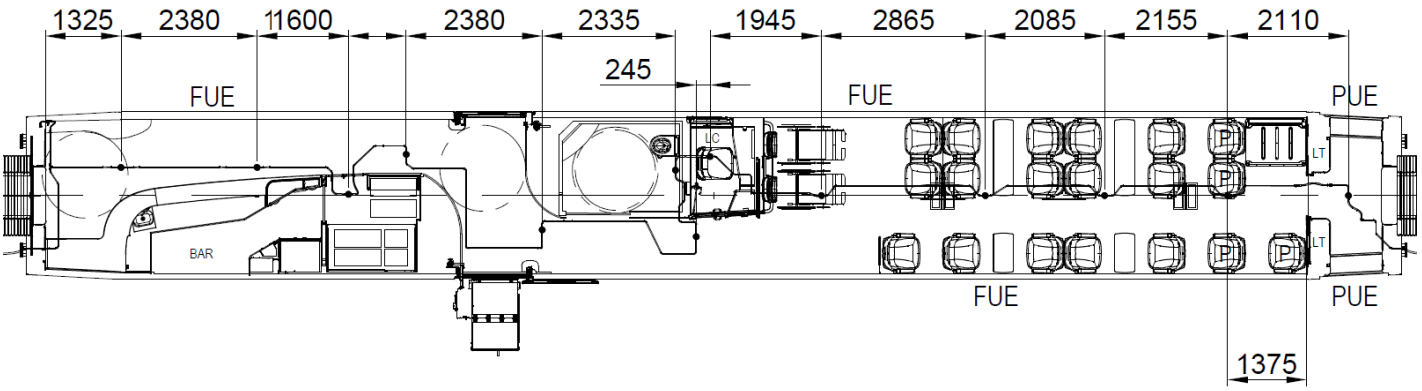
Coach DM1



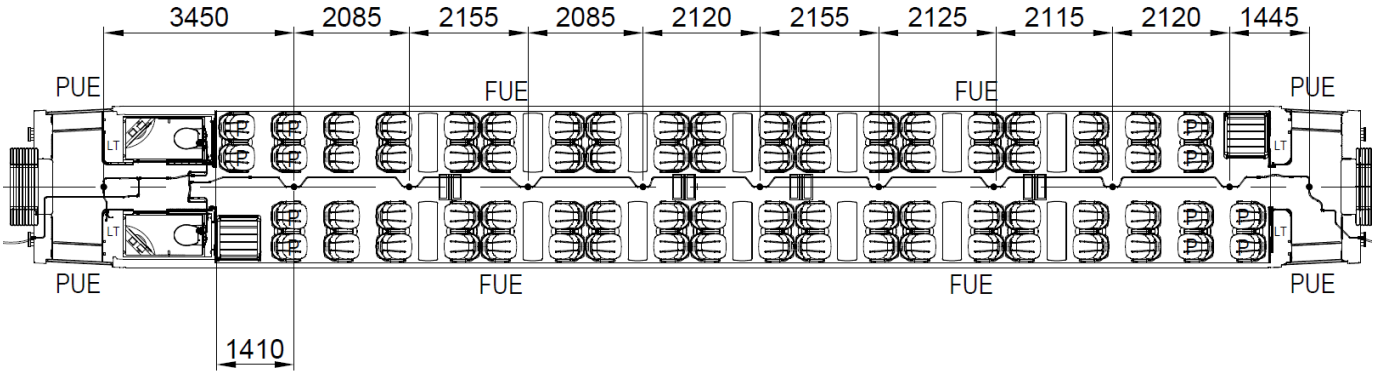
Coach TT2



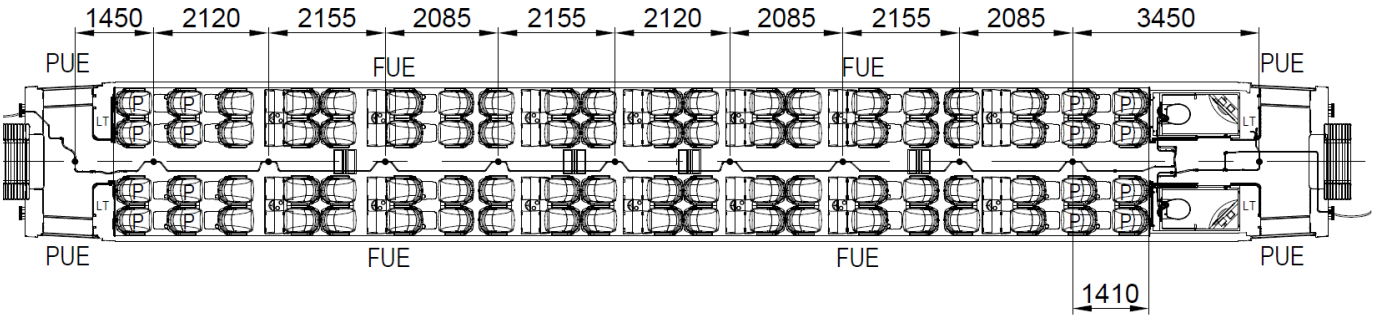
Coach M3



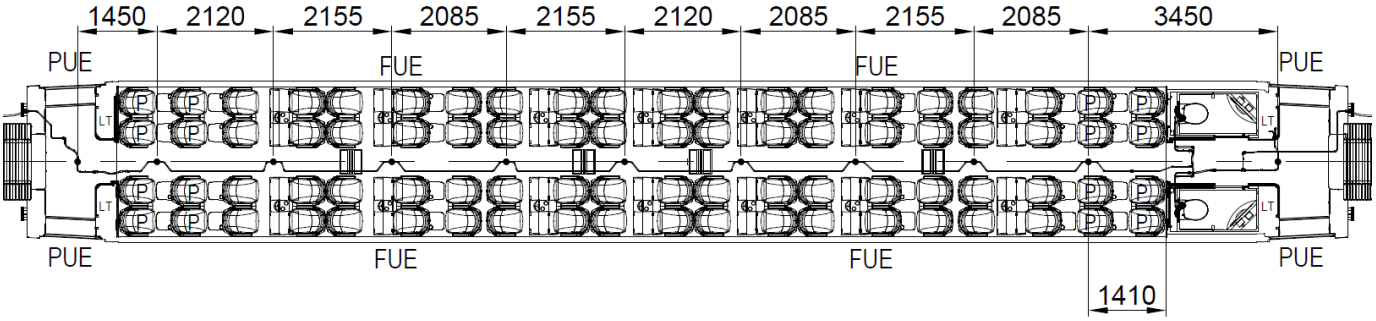
Coach T4



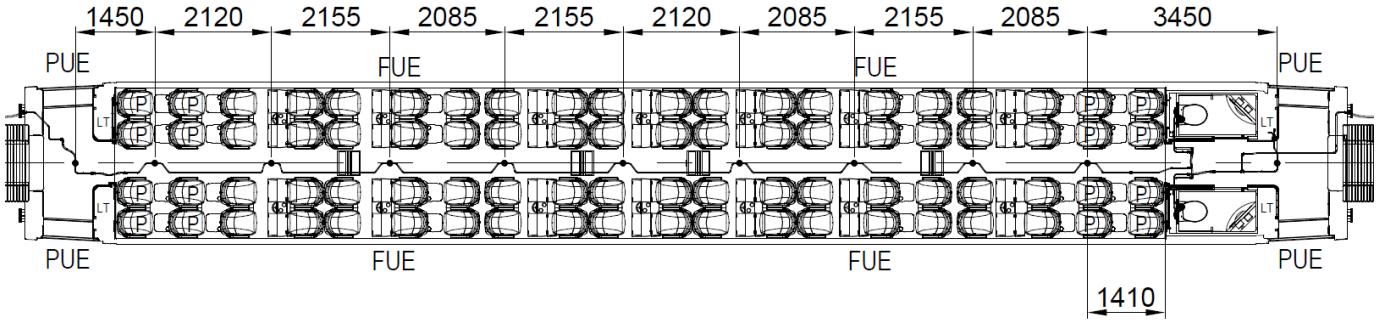
Coach T5



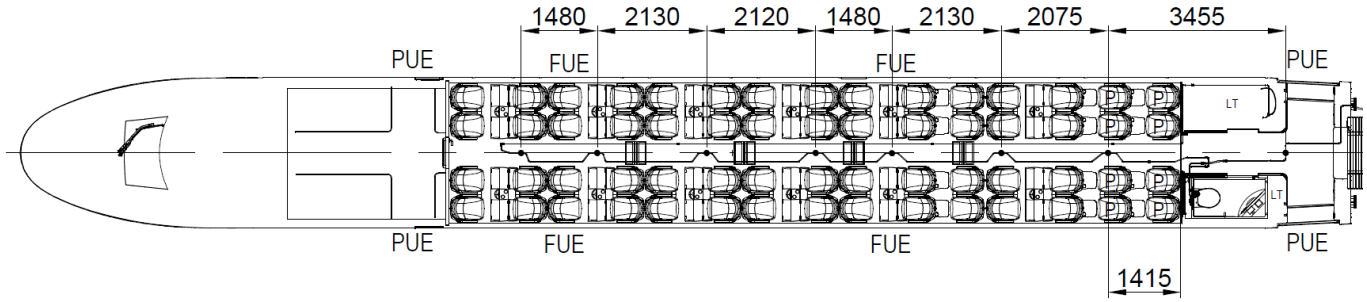
Coach M6



Coach TT7



Coach DM8



3.4.1.2 Technical areas

The fire extinguishing system in the technical areas is performed through aerosol cartridges (FKA) of suitable dimensions installed inside each box where the fire monitoring/extinguishing is necessary. The FPS is able to activate selectively (Extinguishing status) based on the information coming from the detection system and using the RIO I/O modules for the communication.

One RIO rack is able to activate as maximum 2 aerosols at the same time. In case of fire ALARM in a compartment equipped with 3 aerosols, the last one will be activated with a delay time of 5 seconds.

At train level, the FPS is able to activate as maximum 3 aerosols at the same time. If the fire ALARM condition requires the activation of further aerosols, further activation will be provided with a delay time of 5 seconds.

In the following tables are summarized the size needs of the aerosol cartridges (FKA) in order to ensure the correct suppression in each technical compartment in case of fire ALARM:

Car	DM1	TT2				M3	T4
Technical area	**Traction Converters**	**TCB** Transformer Control Box	**MCB** DC Main Circuit Breaker	**CCB** Charging Box	**Distribution Box**	**Traction Converters**	**HV Box**
AEROSOL SIZE FIR.Ex. (FKA)	8x 20g	20g	50g	2x 20g + 50g	50g	8x 20g	100g

Car	T5	M6	TT7				DM8
Technical area	**HV Box**	**Traction Converters**	**TCB** Transformer Control Box	**MCB** DC Main Circuit Breaker	**CCB** Charging Box	**Distribution Box**	**Traction Converters**
AEROSOL SIZE FIR.Ex. (FKA)	100g	8x 20g	20g	50g	2x 20g + 50g	50g	8x 20g

3.4.1.3 Normal condition

In STAND-BY conditions, the piping in the passenger areas is dry.

As soon as the Master FDU detects a fire (ALARM status), the valve DV1 on cylinder unit is commanded to open and the water now pressurized will flow in the main pipe (Pre-activation status). The water mist discharge will happen only through those sprinklers whose heat sensitive bulb has broken due to the high temperature (Suppression status).

As far as the technical areas extinguishing system is concerned, as soon as the Master FDU detects a fire ALARM in a technical area, the aerosol of that area is commanded to open.

Following a fire ALARM in a technical compartment, the FPS system shall:

- Check the connection with the aerosols;
- Check if extinguishing has been activated in the area (accidental or following an ALARM status);
- Command the Shutdown of the Power Supply;
- Proceed with the activation of the aerosol cartriges installed in the technical area subject to fire ALARM.

The activation of the aerosols must take place after a preset delay time from shutting down the Power Supplies in order to ensure that the ventilation fans have completely stopped before dispensing the powder.

3.4.1.4 Failure cases

a) Loss of function of WSU electrovalve (DV1).
b) Breaking of a sprinkler.
c) Loss of piping integrity.
d) Low pressure on the gas cylinder.
e) Low level on the water cylinders.
f) Loss of function of an aerosol activator.

3.4.1.5 Failure detection

a) Electrical continuity of each electrovalve is monitored by the FDU control units and therefore the TCMS is informed.
b) No diagnostic is implemented. Periodic checks are foreseen every 6 months.
c) No diagnostic is implemented. A periodic pipe integrity test is implemented in FPS software and executed every first maintenance slot (**I0**). In case leakages detected after a pipe integrity test, the FDU control units will rise a MAJOR fault and the FAULT lamp on driver's desk will be switched on. The TCMS will be always informed about the results of the piping integrity test.
d) Pressure level in the gas cylinders is monitored by the FDU control units and therefore the TCMS is informed.
e) Periodic checks though a multimeter (once a year) are foreseen.
f) Each aerosol activator is monitored by the control units and therefore the TCMS is informed.

3.4.1.6 Failure handling

a) The function is lost and the driver shall act according the operational manual.
b) Function is endangered. Attended during maintenance as soon as discovered.
c) Function is endangered. Attended during maintenance as soon as discovered.
d) The function is lost and the driver shall act according the operational manual.
e) The function is lost and the driver shall act according the operational manual.
f) The function is lost and the driver shall act according the operational manual.

3.4.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.93
Description	When smoke/heat is detected the fire extinguisher for that specific area shall be activated.
Safety level	SIL 1

Following electrovalves opening, the system is in pre-activated status, i.e. pressurized water flows in all pipes.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.86	As soon as the fire detection system detects a fire in a technical area, the aerosol cartridge installed in that area shall be commanded to open after a suitable delay time (TBD). If the technical area where a fire was detected is equipped with more that one aerosol generator, the FPS shall activate them simultaneously.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_0y.0z.Zefiro-Europe.CONCEPT.HR43	If the technical area where the fire was detected does not have any aerosol generators, the FPS system shall implement all the mitigation actions foreseen for a technical compartment in alarm, but no aerosol generator will be activated.	New Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.230	The information that an aerosol cartridge has been commanded to discharge shall be transmitted to TCMS (Extinguishing status).	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.231	The information of which aerosol cartridge has been commanded to discharge shall be transmitted to TCMS by setting the status of that cartridge to Activated.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.233	As soon as the fire detection system detects a fire in passenger areas, the electrovalve DV1 on WSU unit shall be commanded to open.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_0y.0z.Zefiro-Europe.CONCEPT.HR80	The nitrogen cylinder electrovalve (DV1) must be of "total release" type: once opened it remains in this state till a manual intervention. A loss of power supply shall not cause the closing of the DV1 valve.	New Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.234	DELETED	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.236	The system shall transmit to TCMS the status of Suppression as soon as the WSU DV1 electrovalve is open and the relevant pressure which is detected in the piping trough the transducer MPT.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR21	Passenger compartments shall be automatically protected by nozzles spaced no more than 4 linear meters (maximum 2m from the wall) and will be divided into several independent areas: - Passenger lounges; - Vestibules; - Bistro. Only one nozzle shall be provided inside each toilet. Inside the driver cabs no nozzles are present.	New Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.94
Description	The fire extinguisher shall not be activated for areas were no smoke/heat is detected.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.87	Aerosol generators installed in technical areas where no fire is detected shall not be activated.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.237	Sprinklers shall be provided with a thermal sensitive bulb that breaks as the temperature reaches a certain threshold.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.95
Description	Fire extinguishing system shall perform a continuous self monitoring in order to detect possible failures such as: loss of gas/water in cylinders, incorrect position of valves, etc..
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.243	Electric actuator of each aerosol cartridges shall be monitored.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.244	In case electrical continuity of the actuator is not found, the status of the corresponding cartridge shall be set to Fault .	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR24	During train's service, the TCMS shall execute the alogorithm indicated on par. 3.4.1.1.5 in order to supply compressed air to the suppression conduct.	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR63	In order to supply the compressed air to the suppression conduct, the TCMS sends to the FPS the following commands: -TDV valve open command (CTDVCommand==0) -TDV valve close command (CTDVCommand==1)	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR64	In order to command the TDV valve during the train service based on the alogorithm indicated on par. 3.4.1.1.5, the TCMS shall foreseen the following timers (to be defined): -T0: opening time of the TDV; -TC:TDV opening interval;	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR65	In order to proceed with the alogorithm to supply the compressed air to the suppression conduct, the TCMS shall evaluate the pressure value of the conduct against the preset value Pmin.	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR26	Insulating of the fire suppression system is required to reach the following design parameters. If the train is not in operation and parked without any heating, the system remains ready for the operation (no freezinbg of water) for at least 12 hours in case of inside starting temperature of +20 °C and train minimum temperature will be not less than -10 °C.	New Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_0y.0z.Zefiro-Europe.CONCEPT.HR32	Following a return from a previous ALARM condition without the opening of the nozzles (system in STAND-BY state, but in PRE-ACTIVATION condition), the pipes will be with water under pressure. In this case the FPS proceeds in STAND-BY state for a period not exceeding 48 hours from the Pre-Activation, Then the FPS shall declare a MAJOR Fault.	New Requirement	4S_06.05.-.Fire Protection Systems	EVENT	SIL 2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR33	The pressure level inside the Nitrogen cylinder shall be countinuosly monitored through a pressure transducer (4-20 mA). During all period of the water discharge, on WSU outlet shall be guaranteed a constant pressure, equal or grater the value set on pressure reducer. The pressure value of the reducer shall guarantee on the furthest nozzle the minimum pressure measured during the mock-up test phase.	New Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_0y.0z.Zefiro-Europe.CONCEPT.HR34	The ambient temperature inside the WSU shall be Monitored through temperature transducer (4-20mA) with an accuracy of ± 2°C. This value will be transmitted to the TCMS to carry out the calculation of the CBM on relation with the nitrogen pressure value.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

3.4.2 Interrupt fire extinguisher

This functionality is NOT applicable to ETR1000 VZI R1.

3.5 Maintenance/Test Mode of fire detection system

3.5.1 Function design

3.5.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.81
Description	When the fire detection system is running in Maintenance/Test mode the TCMS shall not show any extra faults or alarms from the fire detection system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.89	The system shall communicate to TCMS that it is running on Maintenance/Test mode.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.246	TCMS shall not show any extra faults or alarms from the FPS as long as FPS status is Maintenance/Test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.108
Description	It shall be possible to test the fire detection system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.93	It shall be possibile to test the FPS critic components (valves, detectors, pipes integrity).	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_0y.0z.Zefiro-Europe.CONCEPT.HR35	The TCMS shall foreseen a push-button on TOD (CTestModeReq==4) in order to enable the piping integrity test procedure to be execute when the train is in maintenance (I0).	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR25	Once the FPS receive the request to execute the piping integrity test from the TCMS, the FPS will enter in TEST TCMS mode and the related procedure will be totally executed by the FPS based on the algorithms indicated on par. 3.4.1.1.4	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR72	After 20 seconds, the TCMS will send the request (CTestModeReq==1) to FPS in order to stop the Pipe integrity test.	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR73	The FPS receive the stop command for the pipe integrity test, but waits till the end of the procedure.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR74	FPS shal foreseen to send to TCMS the following signal in order to monitor the status of the Pipe integrity test: -No test result foreseen (IPipeTestResult==0); -Test successfully passed (IPipeTestResult==1); -Test not passed (IPipeTestResult==3); - Test not executed because not correct status (IPipeTestResult==4)	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR75	In case of a ‘’Test NOT passed’’, the TCMS will generate events according to the TCMS event list.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR77	In case of a ‘’Test NOT passed’’, it is not possible to reset the fault. The pipe Integrity test procedure shall be repeated with a ‘’ successfully passed’’ result in order to eliminate the fault.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR37	In case of pipe integrity test NOT PASSED, FPS proceeds to switch-on the FAULT trainline.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 2

3.6 System Fault Managment

3.6.1 Function design

FPS communicates to TCMS all its internal failure that are divided in 3 main categories:

- **Minor Faults**: produced when a generic fault which does not compromise the main functions of FPS is detected. In this case, the FPS communicates to TCMS a DIAGNOSTIC message.
- **Major Faults**: produced when a fault which leads to a partial loss of one the main functions of FPS is detected. From this category all the failures related to partial loss of fire detection and suppression over each GU are excluded. In this case, the FPS communicates to TCMS a DEGRADATED/OUT OF ORDER fault, depending on the type of fault.
- **Critical Faults**: produced when the system can't perform at least one of its main function or when it lost the partial capability of fire detection or suppression over one of each GU. In this case, the FPS communicates to TCMS a OUT OF ORDER message.

All the faults are communicated by FPS through Ethernet interface, in addiction DEGRADATED and OUT OF ORDER Faults will also lead the FPS to energize the System's Fault Trainline.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.116	Diagnostics from the supplier shall be stored in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_0y.0z.Zefiro-Europe.CONCEPT.HR22	The system shall check for faults related to the entire fire detection/suppression system based on the Fault Table on paragraph 2.2.2. In case of failures recognize by the FPS, the FPS send to TCMS the related Fault message to TCMS.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR38	A MINOR FAULT has an impact on functionality without causing disruptions (loss of redundancies or elements not essential for the fulfillment of functionality). If a MINOR FAULT is detected, the FPS shall: • Communicate a DIAGNOSTIC message in according to TCMS Event Report via TRDP. • Communicate the detail of failure.	New Requirement	4S_06.05.-.Fire Protection Systems	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR68	TCMS shall store an Event (Maintenance) in case of a DIAGNOSTIC received from FPS in according to TCMS Event Report.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR39	A MAJOR FAULT leads to a partial loss of one of main functionalities of the FPS system). If a MAJOR FAULT is detected, the FPS shall: • Communicate DEGRADATED/OUT OF ORDER fault (depending on the type of fault) in according to TCMS Event Report via TRDP. • Energise the FAULT trainline • Communicate the details of the fault.	New Requirement	4S_06.05.-.Fire Protection Systems	EVENT	SIL 2
2F_0y.0z.Zefiro-Europe.CONCEPT.HR69	TCMS shall store an Event (Driver) in case of a DEGRADATED/OUT OF ORDER fault (depending on the type of fault) in according to TCMS Event Report. In case of multiple operation, the message shall be shown on TOD of the enabled driver's desk, specifying what is the unit involved.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR40	A CRITICAL FAULT has a severe impact on the total functionality of the system. If a CRITICAL FAULT is detected, the FPS shall: • Communicate OUT OF ORDER fault in according to TCMS Event Report via TRDP. • Energise the FAULT trainline. • Communicate the details of the fault.	New Requirement	4S_06.05.-.Fire Protection Systems		SIL 2

2F_0y.0z.Zefiro-Europe.CONCEPT.HR70	TCMS shall store an Event (Driver) in case of a OUT OF ORDER fault in according to TCMS Event Report. In case of multiple operation, the message shall be shown on TOD of the enabled driver's desk, specifying what is the unit involved.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR71	In case of MAJOR/CRITICAL fault, following the FAULT trainline energized, the FPS shall switch-on the FAULT lamp on Driver's desk.	New Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

3.7 Events

3.7.1 Function design

3.7.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.113
Description	Diagnostics from the supplier shall be stored in the train diagnostic system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.116	Diagnostics from the supplier shall be stored in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_0y.0z.Zefiro-Europe.CONCEPT.HR46	In the event of a fire PRE-ALARM, the TOD shall store a synoptic with the following information: -the coach and the area where the PRE-ALARM was detected; - the sensor in PRE-ALARM. - In case of multiple running, the above information shall be shown on TOD of the enabled driver's desk, specifying what is the unit involved with the PRE-ALARM.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_0y.0z.Zefiro-Europe.CONCEPT.HR47	In the event of a fire ALARM, the TOD shall store a synoptic with the following information: -the coach and the area where the ALARM occurred; -the sensor or the LHC cable in ALARM; -the state of the extinguishing system; In case of multiple operation, the above information must be shown on the TOD of the enabled driver's desk, what is the unit involved with the ALARM.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_0y.0z.Zefiro-Europe.CONCEPT.HR48	The TOD shall store the information of coach in PRE-ALARM and coach in ALARM status in a different way (e.g. different colour). In case of multiple operation, the above information shall be shown on TOD of the enabled driver's desk specifying specifying what is the unit involved.	New Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

4. Signal Interface

Refer to the documents EC130021300 ‘‘ICD Requirements for FDU’’ and EC130021301 ‘‘ICD Requirements for FDU (Annex)’’.

5. References

[1] EC010019570 - Fire Detection System SRD.

[2] EC010019571 - Fire Suppression System SRD.

[3] EC130021300 - ICD Requirements for FDU.

[4] EC130021301 - ICD Requirements for FDU (Annex).